



Adaptive Gaussian Mixture Filtering for Multi-sensor Maneuvering Cislunar Space Object Tracking

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Abstract

Successful space domain awareness (SDA) requires maintaining track custody of cooperative and noncooperative cislunar space objects (CSOs) through both ballistic and maneuvering trajectories. The surveillance of CSOs is particularly challenging due to the underlying chaotic multi-body dynamics, which makes uncertainty propagation more difficult when compared to Keplerian orbits. While methods exist for tracking cooperative spacecraft using high accuracy range measurements, the problem of passive noncooperative maneuvering CSO tracking has received considerably less attention. In this paper, CSO motion is modeled as a jump Markov system (JMS), where the CSO modality is unknown and subject to random switching. A novel adaptive Bayesian filter is proposed and shown to successfully maintain CSO track custody through both ballistic and maneuvering phases of an Artemis I-like trajectory.

Keywords Cislunar space · Space domain awareness · Maneuvering object tracking · Nonlinear estimation · Negative information · Random finite sets

1 Introduction

Accurate knowledge of a space object's orbit and its associated uncertainty is essential to cislunar space domain awareness (SDA). The probabilistic approach to cislunar space object (CSO) tracking involves finding complete statistical descriptions of state uncertainty conditioned on available noisy observations. Uncertainty realism in cislunar SDA is critical not only for state estimation accuracy, but for downstream objectives including conjunction analysis [1], object search [2], and multi-sensor

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fusion [3]. Tracking a CSO with sparse observations presents significant estimation challenges due in part to the underlying nonlinear, chaotic multi-body dynamics, which result in highly non-Gaussian state distributions [4]. Notional cislunar space-based observer constellations [5] increase observation opportunities yet are still limited by lunar occlusion and albedo, which prevent consistent measurement opportunities, specifically during close approaches to the Moon. During observation gaps, CSO state uncertainty grows rapidly and exhibits a highly non-Gaussian structure driven by the underlying nonlinear, chaotic multi-body dynamics and small non-conservative disturbances, even when the CSO is assumed to be ballistic. Tracking a maneuvering CSO is even more challenging, as the unknown time and magnitude of CSO maneuvers accelerate uncertainty growth.

In the Bayesian approach to state estimation and tracking, full statistical descriptions of the target uncertainty are sought, typically in the form of probability density functions (pdfs). In nonlinear settings, Bayesian Gaussian mixture (GM) and particle filters offer distinct advantages in that their underlying parameterizations can accurately approximate arbitrary non-Gaussian pdfs. Adaptive Gaussian mixture (AGM) filter approaches [6–12] systematically increase and decrease the mixture resolution in key regions of the distribution support to account for strong nonlinear effects while maintaining overall efficiency. Applications of AGM filtering to near-Earth and CSO tracking are demonstrated in [8, 13] and [4], respectively. Other notable estimation approaches for nonlinear systems include, for example, polynomial chaos expansion [14, 15], sigma point [1, 16], Gauss–Hermite quadrature [17], multi-fidelity [18], and GM particle filtering methods [19–21].

The challenging problem of maneuvering target tracking is an extensively studied problem with applications in autonomous vehicle tracking [22, 23], air traffic control [24, 25], and missile defense [26, 27]. Maneuvering objects are commonly modeled as jump Markov systems (JMSs), wherein an object's modality evolves according to a homogeneous Markov chain [28–34]. Each modality consists of its own dynamics and measurement model, and thus, can capture different object behaviors such as ballistic flight and thrusting. The Bayes-optimal filter solution requires exponential complexity to account for the possible target mode histories, motivating suboptimal approximations such as the widely adopted interacting multiple model (IMM) filter [29]. Non-JMS formulations include a control-theoretic approach that detects maneuvers based on the integrated control effort required to connect two boundary states under an assumed optimal control policy [35–39]. Another approach models target maneuvers using a continuous parameter vector that either remains constant or changes according to a Bernoulli trial with known probability [40]. While maneuvering target tracking and Bayesian nonlinear estimation have received extensive attention independently, comparatively little existing work has been dedicated to probabilistic nonlinear estimation of maneuvering objects in chaotic systems.

This article presents the adaptive Gaussian mixture interacting multiple model (AGMIMM) filter, a novel AGM filter for tracking noncooperative maneuvering objects in chaotic dynamical systems. The AGMIMM filter features a multi-pronged adaptation approach. Unlike existing nonlinear AGM estimators, the AGMIMM adaptively accounts for stochastic target modality changes and, unlike existing maneuvering target trackers, employs a comprehensive multi-criterion mixture

adaptation for other important nonlinear effects. In particular, this work considers adaptation criteria designed for capturing nonlinear effects from dynamics, measurements, negative information, and known constraints. Another novel contribution of this work is the adoption of a cluster-based merging algorithm to accurately approximate the entire target mode history, which is shown to significantly improve state estimation performance and robustness compared to existing techniques. The application to multi-sensor noncooperative CSO tracking is then considered, where the maneuvering CSO modality is unknown and subject to random switching. The effectiveness of the new filter is demonstrated in a challenging angles-only tracking problem, where a noncooperative CSO performs multiple unknown maneuvers to achieve an Artemis I-like trajectory. Synthetic angle measurements are generated based on a notional network of cislunar space-based observers flying in a 1:1 resonant orbit configuration.

The remainder of this paper is organized as follows. The problem formulation is described in Sect. 2. Methodology is discussed in Sect. 3 and the AGMIMM algorithm is presented for a general nonlinear system. The application of the AGMIMM filter to the problem of tracking a maneuvering CSO using angles-only data from space-based observers is presented in Sect. 4. Numerical results are presented in Sect. 5 and concluding remarks follow in Sect. 6.

2 Problem Formulation

This paper considers the problem of estimating the state and modality of a noncooperative CSO conditioned on all available sensor measurements. Particularly, the CSO state $\mathbf{x}_k \in \mathbb{R}^{n_x}$ is defined as

$$\mathbf{x}_k = [\mathbf{r}_k^\top \mathbf{v}_k^\top]^\top = [x_k \ y_k \ z_k \ \dot{x}_k \ \dot{y}_k \ \dot{z}_k]^\top, \quad (1)$$

where $(\cdot)_k$ describes the quantities at discrete time t_k , $\mathbf{r}_k \in \mathbb{R}^3$ is the relative position of the target with respect to the Earth–Moon barycenter in the synodic frame, and $\mathbf{v}_k \in \mathbb{R}^3$ is the relative velocity of the target in the synodic frame. Without loss of generality, this paper adopts the circular restricted three-body problem (CR3BP) assumptions. Units of length and time are normalized to ensure that position and velocity components are of the same magnitude. The resulting nondimensional length unit [LU] and time unit [TU] for the considered Earth–Moon system are

$$1 \text{ [LU]} = 379,632.95 \text{ [km]}, \quad (2)$$

$$1 \text{ [TU]} = 368,232.68 \text{ [s]}. \quad (3)$$

Let $\mu = 0.0122$ represent the ratio of the mass of the Moon to the sum of the masses of the Moon and Earth, and let r_{Earth} and r_{Moon} be the distances of the target from the Earth and Moon, respectively. The positions of the primaries in the synodic frame are constant and defined as

$$\mathbf{r}_{\text{Earth}} = [-\mu \ 0 \ 0]^\top \text{ [LU]}, \quad (4)$$

$$\mathbf{r}_{\text{Moon}} = [1 - \mu \ 0 \ 0]^\top \text{ [LU]}. \quad (5)$$

Letting τ denote the target mode, the dynamics of the target are described by the continuous-time differential equations

$$\ddot{x} - 2\dot{y} = w_x(t, \tau) + \frac{\partial U}{\partial x} \Big|_{\mathbf{r}}, \quad \ddot{y} + 2\dot{x} = w_y(t, \tau) + \frac{\partial U}{\partial y} \Big|_{\mathbf{r}}, \quad \ddot{z} = w_z(t, \tau) + \frac{\partial U}{\partial z} \Big|_{\mathbf{r}}, \quad (6)$$

$$U(\mathbf{r}) = \frac{1 - \mu}{r_{\text{Earth}}} + \frac{\mu}{r_{\text{Moon}}} + \frac{x^2 + y^2}{2}, \quad (7)$$

where U is the pseudo-potential and $\mathbf{w}(t, \tau) = [w_x(t, \tau) \ w_y(t, \tau) \ w_z(t, \tau)]^\top$ is the mode-dependent relative acceleration due to unknown disturbances and unknown inputs such as target thrust. If $\mathbf{w}(t, \tau) = \mathbf{0}$, then the classical CR3BP equations of motion (EOMs) are recovered and admit an integral of motion known as the Jacobi constant defined as

$$C(\mathbf{r}_k, v_k) = 2U(\mathbf{r}_k) - v_k^2, \quad (8)$$

where $v_k = \|\mathbf{v}_k\|$.

This paper assumes that target mode changes (e.g., from ballistic to thrusting) occur at discrete time steps and thus form a sequence $(\tau_0, \tau_1, \dots, \tau_k)$. Because the modes are unknown *a priori*, they are modeled as a homogeneous Markov chain with known mode transition probabilities

$$\pi_{ij} = \Pr(\tau_k = j | \tau_{k-1} = i) \quad (9)$$

that are time-invariant and independent of the base state. While the standard assumptions of known mode transition probabilities, time-invariance, and base state independence are necessary for this paper's theoretical developments, the AGMIMM is shown to perform well when all three assumptions are violated in Sect. 5. System modes can take values from the set $\mathcal{M} = \{1, \dots, M\} \subset \mathbb{N}$ where M is the number of possible modes. The target mode τ_k is defined to be left-continuous such that mode τ_k is in effect over $t \in [t_{k-1}^+, t_k^-]$.

Angles-only measurements of the target are taken by a network of O space-based observers each endowed with a body frame

$$\mathcal{B}^{(o)} = \left(\mathbf{r}_{\text{obs}}^{(o)}, \hat{\mathbf{b}}_1^{(o)}, \hat{\mathbf{b}}_2^{(o)}, \hat{\mathbf{b}}_3^{(o)} \right), \quad (10)$$

where $\mathbf{r}_{\text{obs}}^{(o)}$ is the position of observer o with respect to the Earth–Moon barycenter in the synodic frame and $\hat{\mathbf{b}}_1^{(o)}$, $\hat{\mathbf{b}}_2^{(o)}$, and $\hat{\mathbf{b}}_3^{(o)}$ are unit vectors that form an orthogonal basis. Each observer is equipped with a camera whose boresight coincides with the $\hat{\mathbf{b}}_2^{(o)}$ direction. The transformations from the synodic frame to each observer body

frame are assumed known. The observer state $\xi^{(o)}$ consists of observer o 's relative position and orientation and determines the bounded sensor field-of-view (FoV) $\mathcal{S}^{(o)}(\xi^{(o)})$, where $\mathcal{S}^{(o)}$ is taken to be a compact and bounded set in $\mathbb{R}^3$. Observer o is capable of measuring the target if and only if the target is within $\mathcal{S}^{(o)}$, illuminated, not occluded by the Earth/Moon, and not in the Moon exclusion cone (see Fig. 1). Then, the probability of detection for each observer is defined as

$$p_{D,k}^{(o)}(\mathbf{x}; \mathcal{S}^{(o)}) = 1_{\mathcal{S}^{(o)}}(\mathbf{r}) \times \psi(\mathbf{x}, \xi^{(o)}), \quad (11)$$

where the indicator function

$$1_B(\mathbf{a}) \triangleq \begin{cases} 1 & \mathbf{a} \in B, \\ 0 & \text{otherwise,} \end{cases} \quad (12)$$

and $\psi(\mathbf{x}, \xi^{(o)})$ is unity if the target is illuminated, non-occluded, and outside of the Moon exclusion cone for observer o and zero otherwise.

Angular measurements come in the form of azimuth θ and elevation ϕ angles. For each observer, these measurements are concatenated into the vector $[\theta^{(o)} \ \phi^{(o)}]^\top$ defined by the nonlinear map

$$\mathbf{h}^{(o)}(\mathbf{x}) = \begin{bmatrix} \arctan \left(\frac{\hat{\mathbf{u}}^{(o)} \cdot \hat{\mathbf{b}}_1^{(o)}}{\hat{\mathbf{u}}^{(o)} \cdot \hat{\mathbf{b}}_2^{(o)}} \right) \\ \arcsin \left(\hat{\mathbf{u}}^{(o)} \cdot \hat{\mathbf{b}}_3^{(o)} \right) \end{bmatrix}, \quad (13)$$

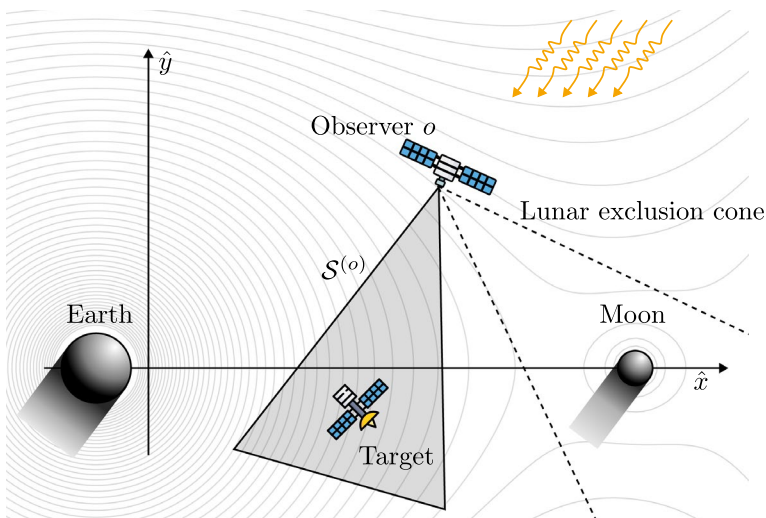


Fig. 1 Example configuration where the target is within the bounded sensor FoV, illuminated, not occluded by the Earth/Moon, and not in the Moon exclusion angle where gray lines denote contour lines of U (note Earth, Moon, target, and observer are not to scale)

where

$$\hat{\mathbf{u}}^{(o)} = \frac{\mathbf{r} - \mathbf{r}_{\text{obs}}^{(o)}}{\|\mathbf{r} - \mathbf{r}_{\text{obs}}^{(o)}\|}. \quad (14)$$

At each observation time k , observer o produces a Bernoulli random finite set (RFS) measurement $Z_k^{(o)}$ that is either empty or contains a single vector measurement $\mathbf{z}_k^{(o)}$, depending on the probability of detection. The multi-sensor measurement Z_k is then defined as the tuple

$$Z_k \triangleq (Z_k^{(1)}, \dots, Z_k^{(O)}). \quad (15)$$

This paper assumes that the continuous-time dynamics can be accurately approximated as a discrete-time nonlinear JMS

$$\mathbf{x}_k = \mathbf{f}(\mathbf{x}_{k-1}) + \mathbf{w}_{k-1}(\tau_k), \quad (16)$$

$$\mathbf{z}_k^{(o)} = \mathbf{h}^{(o)}(\mathbf{x}_k) + \mathbf{v}_k(\tau_k), \quad (17)$$

where $\mathbf{f}(\mathbf{x})$ is defined as the solution flow of the CR3BP EOMs and the process noise $\mathbf{w}_{k-1}(\tau_k)$ and measurement noise $\mathbf{v}_k(\tau_k)$ are zero-mean, Gaussian, and white with covariances $E[\mathbf{w}_{k-1}(\tau_k)\mathbf{w}_{k-1}(\tau_k)^\top] = \mathbf{Q}_{k-1}(\tau_k)$ and $E[\mathbf{v}_k(\tau_k)\mathbf{v}_k(\tau_k)^\top] = \mathbf{R}_k(\tau_k)$, where $E[\cdot]$ is the expectation operator. The target kinematic state (also known as the base state) and modality are correlated and thus modeled as the joint state $\tilde{\mathbf{x}}_k = [\mathbf{x}_k^\top \tau_k]^\top$. With assumed knowledge of the pdf $p_0(\tilde{\mathbf{x}}_0)$ at some time t_0 , this paper considers the problem of recursively estimating the unknown prior and posterior conditional distributions, $p_{k|k-1}(\tilde{\mathbf{x}}_k|Z_{0:k-1})$ and $p_k(\tilde{\mathbf{x}}_k|Z_{0:k})$, respectively, where $Z_{0:k}$ denotes the entire measurement history from t_0 to t_k . The evolution of these densities follows

$$\cdots \rightarrow p_{k-1}(\tilde{\mathbf{x}}_{k-1}|Z_{0:k-1}) \rightarrow p_{k|k-1}(\tilde{\mathbf{x}}_k|Z_{0:k-1}) \rightarrow p_k(\tilde{\mathbf{x}}_k|Z_{0:k}) \rightarrow \cdots \quad (18)$$

3 Methodology

This paper presents a novel adaptive Bayesian filter that is designed to handle the challenging angles-only tracking problem of a maneuvering target in cislunar space in the presence of long observation gaps. Such a problem exhibits strong nonlinearities in both the dynamics and measurement functions which result in the evolution of highly non-Gaussian distributions. As such, typical central moment filters prove inadequate for robust and accurate CSO target tracking in these settings [4, 41]. The challenging density evolution is exacerbated further with the introduction of maneuvers with unknown ignition time, duration, and magnitude. This paper presents the AGMIMM where pdfs take the form of GMs to accurately represent the true non-Gaussian distributions. Furthermore, the proposed filter is classified as an AGM filter wherein automatic component splitting increases GM resolution in key areas

to account for chaotic dynamics, nonlinear measurements, bounded sensor FoVs, and other nonlinear constraints. The AGMIMM utilizes the foundation of the IMM which makes approximations of the exact nonlinear JMS filter to maintain computational tractability [29].

3.1 Nonlinear JMS Bayes Filter

The problem of target tracking when the system is modeled as a nonlinear JMS requires the estimation of both the target's state and modality. Let $g_k(Z_k|\tilde{\mathbf{x}})$ denote the joint multi-sensor likelihood function. Given the multi-sensor measurement Z_k and prior $p_{k|k-1}(\tilde{\mathbf{x}})$, the posterior distribution is given by Bayes' rule as

通过先前的观测预测下一时刻的状态

$$\text{后验} \quad p_{k|k}(\tilde{\mathbf{x}}|Z_{0:k}) = \frac{g_k(Z_k|\tilde{\mathbf{x}})p_{k|k-1}(\tilde{\mathbf{x}})}{\int \sum_{\ell=1}^M g_k(Z_k|\tilde{\mathbf{x}}^{(\ell)})p(\tilde{\mathbf{x}}^{(\ell)})d\mathbf{x}}, \quad (19)$$

where the notation $\tilde{\mathbf{x}}^{(j)} = [\mathbf{x}_k^\top \tau_k = j]^\top$ exposes the modality value. Assuming that measurements are conditionally independent of the target kinematic state and mode, the joint multi-sensor measurement likelihood is given by the product

$$g_k(Z_k|\tilde{\mathbf{x}}) = \prod_{o=1}^O g_k^{(o)}(Z_k^{(o)}|\tilde{\mathbf{x}}), \quad (20)$$

where $g_k^{(o)}(Z_k^{(o)}|\tilde{\mathbf{x}})$ is the measurement likelihood function for observer o . By this factorization, the multi-sensor Bayes' update can be written as the composition of single-sensor updates as

$$p_k(\tilde{\mathbf{x}}_k^{(j)}|Z_{0:k}) = \Psi_k^{(O)} \circ \dots \circ \Psi_k^{(1)} p_{k|k-1}(\tilde{\mathbf{x}}_k^{(j)}|Z_{0:k-1}) \quad (21)$$

as shown in [42], where the single-sensor Bayes update operator $\Psi_k^{(o)}$ is

$$[\Psi_k^{(o)} p](\tilde{\mathbf{x}}) = \frac{g_k^{(o)}(Z_k^{(o)}|\tilde{\mathbf{x}})p(\tilde{\mathbf{x}})}{\int \sum_{\ell=1}^M g_k^{(o)}(Z_k^{(o)}|\tilde{\mathbf{x}}^{(\ell)})p(\tilde{\mathbf{x}}^{(\ell)})d\mathbf{x}}. \quad (22)$$

The filter prediction step is a direct result of the Chapman–Kolmogorov equation and total probability theorem and is given by

先验

$$p_{k|k-1}(\tilde{\mathbf{x}}_k^{(j)}|Z_{0:k-1}) = \int \sum_{i=1}^M p_{k|k-1}(\tilde{\mathbf{x}}_k^{(j)}|\tilde{\mathbf{x}}_{k-1}^{(i)})p_{k-1}(\tilde{\mathbf{x}}_{k-1}^{(i)}|Z_{0:k-1})d\mathbf{x}_{k-1}, \quad (23)$$

where the joint state transition density $p_{k|k-1}(\tilde{\mathbf{x}}_k^{(j)}|\tilde{\mathbf{x}}_{k-1}^{(i)})$ is known and specified by the mode transition probabilities, dynamics function, and process noise model. Closed-form solutions of (21) and (23) are available when $p_0(\mathbf{x}_0)$ is Gaussian and the state transition and measurement likelihood are linear-Gaussian, yet require $\mathcal{O}(M^k)$ terms. Together, the Markov chain-driven exponential complexity, nonlinear dynamics, and

nonlinear observations considered in this paper necessitate suboptimal approximations for tractability.

The joint distribution is readily factored into the mode-conditioned densities and mode probabilities. The prior and posterior pdfs of $\mathbf{x}_k$ conditioned on the j th system mode are defined as

$$\text{先验} \quad p_{k|k-1}^{(j)}(\mathbf{x}_k) = p_{k|k-1}(\mathbf{x}_k | \tau_k = j, Z_{0:k-1}), \quad (24)$$

$$\text{后验} \quad p_k^{(j)}(\mathbf{x}_k) = p_k(\mathbf{x}_k | \tau_k = j, Z_{0:k}), \quad (25)$$

where the superscripts denote the assumed system mode. Let $\mu_{k-1}^{(i)}$ denote the posterior probability of mode i at time t_{k-1} defined as

$$\text{后验} \quad \mu_{k-1}^{(i)} = \Pr(\tau_{k-1} = i | Z_{0:k-1}) \quad (26)$$

and $\bar{\mu}_{k|k-1}^{(j)}$ denote the prior probability of mode j at time t_k defined as

$$\text{先验} \quad \bar{\mu}_{k|k-1}^{(j)} = \Pr(\tau_k = j | Z_{0:k-1}) = \sum_{i=1}^M \pi_{ij} \mu_{k-1}^{(i)}. \quad (27)$$

The single sensor update operator applied to an arbitrary mode-conditioned density $p^{(j)}(\mathbf{x})$ and prior mode probability $\bar{\mu}^{(j)}$ is defined as

$$\left[\Psi_k^{(o,j)} p^{(j)} \right](\mathbf{x}) = \frac{g_k^{(o,j)}(Z_k^{(o)} | \mathbf{x}) p^{(j)}(\mathbf{x})}{\int \sum_{\ell=1}^M g_k^{(o,\ell)}(Z_k^{(o)} | \mathbf{x}) p^{(\ell)}(\mathbf{x}) d\mathbf{x}}, \quad (28)$$

$$\left[\Psi_k^{(o,j)} \bar{\mu}^{(j)} \right] = \frac{p_k^{(o,j)}(Z_k^{(o)} | Z_{0:k-1}) \bar{\mu}^{(j)}}{\sum_{\ell=1}^M p_k^{(o,\ell)}(Z_k^{(o)} | Z_{0:k-1}) \bar{\mu}^{(\ell)}}, \quad (29)$$

where

$$g_k^{(o,j)}(Z_k^{(o)} | \mathbf{x}) = g_k^{(o)}(Z_k^{(o)} | \tau_k = j, \mathbf{x}), \quad (30)$$

$$p_k^{(o,j)}(Z_k^{(o)} | Z_{0:k-1}) = p_k^{(o)}(Z_k^{(o)} | \tau_k = j, Z_{0:k-1}). \quad (31)$$

Then, the mode-conditioned multi-sensor Bayes' filter equations follow from (23) and (21) and are given by

$$p_{k|k-1}^{(j)}(\mathbf{x}_k) = \int p_{k|k-1}(\mathbf{x}_k | \mathbf{x}_{k-1}, \tau_k = j) p_{k-1}(\mathbf{x}_{k-1} | \tau_k = j, Z_{0:k-1}) d\mathbf{x}_{k-1}, \quad (32)$$

$$p_k^{(j)}(\mathbf{x}_k) = \Psi_k^{(O,j)} \circ \dots \circ \Psi_k^{(1,j)} p_{k|k-1}^{(j)}(\mathbf{x}_k | Z_{0:k-1}), \quad (33)$$

where $p_{k-1}(\mathbf{x}_{k-1}|\tau_k = j, Z_{0:k-1})$ is referred to as the mixed mode-conditioned posterior pdf and is given as

$$p_{k-1}(\mathbf{x}_{k-1}|\tau_k = j, Z_{0:k-1}) = \sum_{i=1}^M \left[\frac{\pi_{ij}\mu_{k-1}^{(i)}}{\bar{\mu}_{k|k-1}^{(j)}} p_{k-1}^{(i)}(\mathbf{x}_{k-1}) \right]. \quad (34)$$

The update of the prior mode probability $\bar{\mu}_{k|k-1}^{(j)}$ is defined as

$$\mu_k^{(j)} = \Psi_k^{(O,j)} \circ \dots \circ \Psi_k^{(1,j)} \bar{\mu}_{k|k-1}^{(j)}. \quad (35)$$

The remainder of this section is organized as follows: Sect. 3.2 presents a novel application of a cluster-based merging algorithm to approximate (34) in order to maintain reasonable computational complexity, Sect. 3.3 discusses the approximations of (32) and (33), and Sect. 3.4 defines the criteria that informs the adaptive splitting techniques.

3.2 Bayesian IMM Approximations

In the application of the recursion defined in (32) and (33), calculation of the mixed mode-conditioned posterior results in a pdf with an exponentially growing number of terms. It is assumed that the posterior pdf $p_{k-1}^{(i)}(\mathbf{x}_{k-1})$ is modeled as the GM

$$p_{k-1}^{(i)}(\mathbf{x}_{k-1}) = \sum_{\ell=1}^{L_{k-1}^{(i)}} \omega_{k-1}^{(\ell,i)} \mathcal{N}(\mathbf{x}_{k-1}; \mathbf{m}_{k-1}^{(\ell,i)}, \mathbf{P}_{k-1}^{(\ell,i)}), \quad (36)$$

where the superscript $(\cdot)^{(\ell,i)}$ denotes the ℓ th component for the assumed mode $\tau_{k-1} = i$. Note that the GM assumption is minimally restrictive, as GMs are universal function approximators for the class of pdfs with a finite number of discontinuities (which are of practical interest) [43]. The resulting mixed mode-conditioned posterior is then a $(\sum_{i=1}^M L_{k-1}^{(i)})$ -component GM

$$p_{k-1}(\mathbf{x}_{k-1}|\tau_k = j, Z_{0:k-1}) = \sum_{i=1}^M \sum_{\ell=1}^{L_{k-1}^{(i)}} \frac{\pi_{ij}\mu_{k-1}^{(i)}}{\bar{\mu}_{k|k-1}^{(j)}} \omega_{k-1}^{(\ell,i)} \mathcal{N}(\mathbf{x}_{k-1}; \mathbf{m}_{k-1}^{(\ell,i)}, \mathbf{P}_{k-1}^{(\ell,i)}). \quad (37)$$

In successive time steps, the number of components of the mixed mode-conditioned posterior grows exponentially necessitating the need for suboptimal approximations to retain computational tractability. At each time step, the mixed mode-conditioned posterior is approximated as

$$p_{k-1}(\mathbf{x}_{k-1}|\tau_k = j, Z_{0:k-1}) \approx \sum_{\ell=1}^L \omega_{k-1|k-1}^{(\ell,j)} \mathcal{N}(\mathbf{x}_{k-1|k-1}; \mathbf{m}_{k-1|k-1}^{(\ell,j)}, \mathbf{P}_{k-1|k-1}^{(\ell,j)}), \quad (38)$$

Fig. 2 Comparison of the merging of the **a** mixed mode-conditioned posterior density under the **b** IMM, **d** GMIMM, and **f** AGMIMM approximations and subsequent ballistic-mode 21-day propagation of the hypothesis-merged mixtures of the **c** IMM, **e** GMIMM, and **g** AGMIMM, where cyan lines denote 1σ curves, red dots denote Monte Carlo samples, and the log color scale represents the value of the pdfs

where $L < \sum_{i=1}^M L_{k-1}^{(i)} = L_{\max}$ and the subscript $(\cdot)_{k-1|k-1}$ and superscript $(\cdot)^{(\ell,j)}$ distinguish the approximate weights and pdfs in (38) from those seen in (37).

The most notable approximation is the IMM, where $L = 1$ and the mode-conditioned pdfs are assumed Gaussian [29]. As such, the mixed mode-conditioned posterior is approximated as

$$p_{k-1}(\mathbf{x}_{k-1} | \tau_k = j, Z_{0:k-1}) \approx \mathcal{N}(\mathbf{x}_{k-1}; \mathbf{m}_{k-1|k-1}^{(j)}, \mathbf{P}_{k-1|k-1}^{(j)}), \quad (39)$$

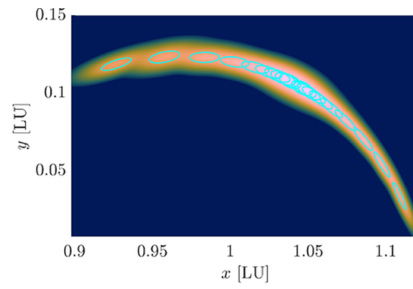
where the mean $\mathbf{m}_{k-1|k-1}^{(j)}$ and covariance $\mathbf{P}_{k-1|k-1}^{(j)}$ are chosen to match the moments of (37). The assumption of Gaussianity is reasonable in some cases but not where highly nonlinear and chaotic dynamics result in highly non-Gaussian distributions (as evidenced by numerical experiments conducted in [44]).

The Gaussian mixture interacting multiple model (GMIMM) generalizes the IMM to account for GM representations of the mode-conditioned pdfs [30]. The mixed mode-conditioned posterior is approximated by retaining the r components with the highest weights and merging the remaining components into a single Gaussian using moment-matching techniques. In other words, $L = r + 1$ and the mixed mode-conditioned posterior is approximated as

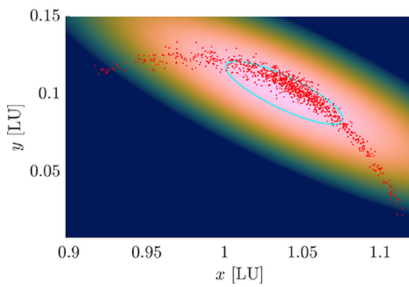
$$p_{k-1}(\mathbf{x}_{k-1} | \tau_k = j, Z_{0:k-1}) \approx \sum_{\ell=1}^{r+1} \omega_{k-1|k-1}^{(\ell,j)} \mathcal{N}(\mathbf{x}_{k-1}; \mathbf{m}_{k-1|k-1}^{(\ell,j)}, \mathbf{P}_{k-1|k-1}^{(\ell,j)}). \quad (40)$$

As evidenced by Fig. 2b, d, both the approximation methods of the IMM and GMIMM fail to accurately model the true mixed mode-conditioned posterior. In both cases, the approximations underestimate the distribution tails while introducing erroneous probability mass in low-probability regions of the support.

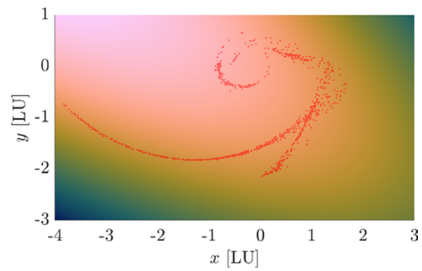
This paper proposes a new approximation of the mixed mode-conditioned posterior based on recent advances in GM reduction methods. Specifically, the AGMIMM filter incorporates the cluster-based merging algorithm of [45, 46] to efficiently approximate the mixed mode-conditioned posterior while minimizing the integrated square distance between the true and approximate density. The Gaussian mixture reduction via clustering (GMRC) merging algorithm consists of a preprocessing, clustering, and an optional refinement step. In the preprocessing step, a preliminary solution to the reduced GM (38) is obtained using Runnalls' merging algorithm [47], where the resulting L components act as initial centers in the following clustering step. Then, a modified k-means algorithm is employed to assign components of (37) to the L cluster centers according to the Kullback–Leibler divergence. The final phase is an optional refinement process where the Newton–Raphson method is used to optimize the component weights, means, and covariances to minimize the normalized integrated squared distance (NISD) between the original



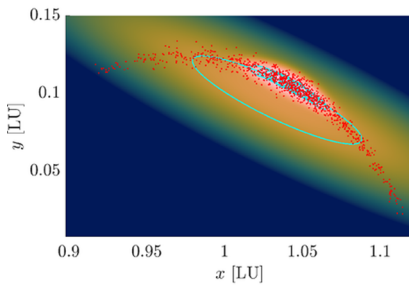
(a)



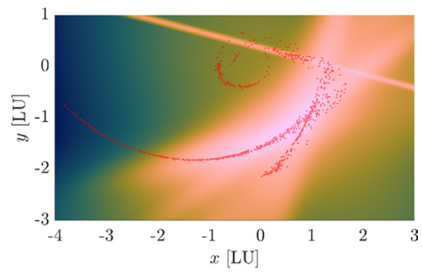
(b)



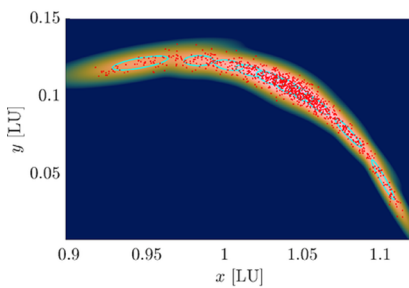
(c)



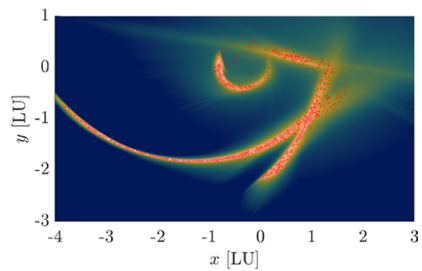
(d)



(e)



(f)



(g)

and merged density. The weights are then reoptimized under the constraint that they must be positive and sum to unity. This paper omits the final refinement step to reduce the computational complexity of the algorithm. An algorithmic description of the GMRC is provided in “Appendix”, and an open-source implementation of the GMRC is available at [48, 49].

By the proposed GMRC reduction, the mixed mode-conditioned posterior (37) is then approximated as

$$p_{k-1}(\mathbf{x}_{k-1} | \tau_k = j, Z_{0:k-1}) \approx \sum_{\ell=1}^L \omega_{k-1|k-1}^{(\ell,j)} \mathcal{N}(\mathbf{x}_{k-1}; \mathbf{m}_{k-1|k-1}^{(\ell,j)}, \mathbf{P}_{k-1|k-1}^{(\ell,j)}), \quad (41)$$

where $L < L_{\max}$ is the desired number of components and the weights, means, and covariances are found via the GMRC algorithm. Like the GMIMM, the AGMIMM approximation approaches the true mode-conditioned posterior as $L \rightarrow L_{\max}$ and, at $L = 1$, is equivalent to the IMM approximation. For $1 < L < L_{\max}$, the GMIMM combines the $L_{\max} - L$ lowest-weight components, ignoring their spatial separation. Thus, the resulting merged component often has a large covariance. Furthermore, when the original distribution is curved or multi-modal, the resulting merged component mean may lie outside the true distribution support, as shown in Fig. 2d. On the other hand, the cluster-based AGMIMM generally provides a more accurate approximation of the true mixed mode-conditioned posterior compared to the GMIMM, as shown in Fig. 2f.

3.3 Gaussian Mixture Filter with State-Dependent Detection

With an adequate approximation of the mixed mode-conditioned posterior, by (32), the mode-conditioned prior at t_k is

$$p_{k|k-1}^{(j)}(\mathbf{x}_k) = \sum_{\ell=1}^{L_{k|k-1}^{(j)}} \omega_{k-1|k-1}^{(\ell,j)} \int \mathcal{N}(\mathbf{x}_k; \mathbf{f}(\mathbf{x}_{k-1}), \mathbf{Q}_{k-1}^{(j)}) \mathcal{N}(\mathbf{x}_{k-1}; \mathbf{m}_{k-1|k-1}^{(\ell,j)}, \mathbf{P}_{k-1|k-1}^{(\ell,j)}) d\mathbf{x}_{k-1}. \quad (42)$$

A closed-form solution of (42) is not available, in general, for systems with non-linear dynamics. For sufficiently small covariances, the mode-conditioned prior pdf can be approximated with moment-matching techniques as the GM [43]

$$p_{k|k-1}^{(j)}(\mathbf{x}_k) \approx \sum_{\ell=1}^{L_{k|k-1}^{(j)}} \omega_{k|k-1}^{(\ell,j)} \mathcal{N}(\mathbf{x}_k; \mathbf{m}_{k|k-1}^{(\ell,j)}, \mathbf{P}_{k|k-1}^{(\ell,j)}), \quad (43)$$

where

$$\omega_{k|k-1}^{(\ell,j)} = \omega_{k-1|k-1}^{(\ell,j)}, \quad (44)$$

$$\mathbf{m}_{k|k-1}^{(\ell,j)} = \int \mathbf{f}(\mathbf{x}_{k-1}) \mathcal{N}(\mathbf{x}_{k-1}; \mathbf{m}_{k-1|k-1}^{(\ell,j)}, \mathbf{P}_{k-1|k-1}^{(\ell,j)}) d\mathbf{x}_{k-1}, \quad (45)$$

$$\mathbf{P}_{k|k-1}^{(\ell,j)} = \int \left(\mathbf{f}(\mathbf{x}_{k-1}) - \mathbf{m}_{k|k-1}^{(\ell,j)} \right) (\cdot)^\top \mathcal{N}(\mathbf{x}_{k-1}; \mathbf{m}_{k-1|k-1}^{(\ell,j)}, \mathbf{P}_{k-1|k-1}^{(\ell,j)}) d\mathbf{x}_{k-1} + \mathbf{Q}_{k-1}^{(j)}, \quad (46)$$

where for brevity, the “ $(\cdot)$ ” in (46) denotes the repetition of the preceding parenthetical quantity. The validity of the small mixand covariance is essential to the accuracy of (43). Specifically, the covariances should be small enough such that scaled variations in $\mathbf{m}_{k-1|k-1}^{(\ell,j)}$ behave linearly over $[t_{k-1}, t_k]$. This requirement provides the central motivation of adaptive GM filtering, as discussed in Sect. 3.4.

Following the prediction step, the mode-conditioned prior is updated iteratively by the single sensor update operator $\Psi_k^{(o,j)}$. Let the mode-conditioned pdf be a GM of the form

$$p^{(j)}(\mathbf{x}) = \sum_{\ell=1}^{L^{(j)}} \omega^{(\ell,j)} \mathcal{N}(\mathbf{x}; \mathbf{m}^{(\ell,j)}, \mathbf{P}^{(\ell,j)}). \quad (47)$$

The single-sensor RFS measurement likelihood is given by

$$g_k^{(o,j)}(Z_k^{(o)} | \mathbf{x}_k) = \begin{cases} 1 - p_{D,k}^{(o)}(\mathbf{x}_k; \mathcal{S}^{(o)}) & Z_k^{(o)} = \emptyset, \\ p_{D,k}^{(o)}(\mathbf{x}_k; \mathcal{S}^{(o)}) g_k^{(o,j)}(\mathbf{z}_k^{(o)} | \mathbf{x}_k) & Z_k^{(o)} = \{\mathbf{z}_k\}, \end{cases} \quad (48)$$

and accounts for both measurement noise and missed detections. Then, by (22), each single-sensor iteration of the multi-sensor Bayes update yields an updated mode-conditioned pdf of the form

$$\begin{aligned} \left[\Psi_k^{(o,j)} p^{(j)} \right](\mathbf{x}) &\propto 1_\emptyset(Z_k^{(o)}) \sum_{\ell=1}^{L^{(j)}} (1 - p_{D,k}(\mathbf{x}; \mathcal{S}^{(o)})) \omega^{(\ell,j)} \mathcal{N}(\mathbf{x}; \mathbf{m}^{(\ell,j)}, \mathbf{P}^{(\ell,j)}) \\ &+ (1 - 1_\emptyset(Z_k^{(o)})) \sum_{\ell=1}^{L^{(j)}} p_{D,k}(\mathbf{x}; \mathcal{S}^{(o)}) \omega^{(\ell,j)} \prod_{\mathbf{z} \in Z_k^{(o)}} g_k^{(o,j)}(\mathbf{z} | \mathbf{x}_k) \mathcal{N}(\mathbf{x}; \mathbf{m}^{(\ell,j)}, \mathbf{P}^{(\ell,j)}), \end{aligned} \quad (49)$$

where the single-sensor measurement likelihood function is $g_k^{(o,j)}(\mathbf{z} | \mathbf{x}_k) = \mathcal{N}(\mathbf{z}; \mathbf{h}^{(o)}(\mathbf{x}_k), \mathbf{R}_k^{(j)})$. Note that, by the RFS-theoretic measurement likelihood and state-dependent probability of detection function, negative information from non-detection is systematically assimilated in a unified Bayes update. On the other hand, the state-dependent probability of detection is nonlinear in $\mathbf{x}$. To that end, a zeroth-order Taylor series expansion can be applied about the means, as shown in [12]. Then, the single sensor update of a mode-conditioned prior is approximated as the GM

$$\left[\Psi_k^{(o,j)} p^{(j)}\right](\mathbf{x}) \approx p_+^{(j)}(\mathbf{x}) = \sum_{\ell=1}^{L_+^{(j)}} \omega_+^{(\ell,j)} \mathcal{N}(\mathbf{x}; \mathbf{m}_+^{(\ell,j)}, \mathbf{P}_+^{(\ell,j)}), \quad (50)$$

where if $Z_k^{(o)} = \{\mathbf{z}_k^{(o)}\}$,

$$\omega_+^{(\ell,j)} \propto \omega^{(\ell,j)} p_{D,k}^{(o)}(\mathbf{m}^{(\ell,j)}; \mathcal{S}^{(o)}) q_+^{(\ell,j)}(\mathbf{z}_k^{(o)}), \quad (51)$$

$$\mathbf{m}_+^{(\ell,j)} = \mathbf{m}^{(\ell,j)} + \mathbf{K}_+^{(\ell,j)}(\mathbf{z}_k^{(o)} - \hat{\mathbf{z}}_+^{(\ell,j)}), \quad (52)$$

$$\mathbf{P}_+^{(\ell,j)} = \mathbf{P}^{(\ell,j)} - \mathbf{C}_+^{(\ell,j)}(\mathbf{K}_+^{(\ell,j)})^\top - \mathbf{K}_+^{(\ell,j)}(\mathbf{C}_+^{(\ell,j)})^\top + \mathbf{K}_+^{(\ell,j)}\mathbf{P}_{z,+}^{(\ell,j)}(\mathbf{K}_+^{(\ell,j)})^\top, \quad (53)$$

$$q_+^{(\ell,j)}(\mathbf{z}_k^{(o)}) = \mathcal{N}(\mathbf{z}_k^{(o)}; \hat{\mathbf{z}}_+^{(\ell,j)}, \mathbf{P}_{z,+}^{(\ell,j)}), \quad (54)$$

$$\hat{\mathbf{z}}_+^{(\ell,j)} = \int \mathbf{h}^{(o)}(\mathbf{x}) \mathcal{N}(\mathbf{x}; \mathbf{m}^{(\ell,j)}, \mathbf{P}^{(\ell,j)}) d\mathbf{x}, \quad (55)$$

$$\mathbf{C}_+^{(\ell,j)} = \int (\mathbf{x} - \mathbf{m}^{(\ell,j)})(\mathbf{h}^{(o)}(\mathbf{x}) - \hat{\mathbf{z}}_+^{(\ell,j)})^\top \mathcal{N}(\mathbf{x}; \mathbf{m}^{(\ell,j)}, \mathbf{P}^{(\ell,j)}) d\mathbf{x}, \quad (56)$$

$$\mathbf{P}_{z,+}^{(\ell,j)} = \int (\mathbf{h}^{(o)}(\mathbf{x}) - \hat{\mathbf{z}}_+^{(\ell,j)})(\cdot)^\top \mathcal{N}(\mathbf{x}; \mathbf{m}^{(\ell,j)}, \mathbf{P}^{(\ell,j)}) d\mathbf{x} + \mathbf{R}_k^{(j)}, \quad (57)$$

$$\mathbf{K}_+^{(\ell,j)} = \mathbf{C}_+^{(\ell,j)}(\mathbf{P}_{z,+}^{(\ell,j)})^{-1}, \quad (58)$$

where the covariance update of (53) is given in an alternative Joseph-stabilized form [50]. Equations (55)–(58) are due to [51]. If $Z_k^{(o)} = \emptyset$, the components of the mode-conditioned posterior are simply

$$\omega_+^{(\ell,j)} \propto \omega^{(\ell,j)}(1 - p_{D,k}^{(o)}(\mathbf{m}^{(\ell,j)}; \mathcal{S}^{(o)})), \quad (59)$$

$$\mathbf{m}_+^{(\ell,j)} = \mathbf{m}^{(\ell,j)}, \quad (60)$$

$$\mathbf{P}_+^{(\ell,j)} = \mathbf{P}^{(\ell,j)}. \quad (61)$$

In addition to the prior mode-conditioned pdf, the prior mode probability $\bar{\mu}_{k|k-1}^{(j)}$ must be updated. Similarly, the single sensor update operator is applied recursively on $\bar{\mu}_{k|k-1}^{(j)}$ to obtain the posterior mode probability at t_k . Let $\bar{\mu}^{(j)}$ be an arbitrary prior mode probability. Starting from (29) and applying (48), a single iteration of the single sensor update operator is

$$\bar{\mu}_+^{(j)} \propto \begin{cases} \bar{\mu}^{(j)} \sum_{\ell=1}^{L_+^{(j)}} \omega^{(\ell,j)} (1 - p_{D,k}^{(o)}(\mathbf{m}^{(\ell,j)}; \mathcal{S}^{(o)})) & Z_k^{(o)} = \emptyset, \\ \bar{\mu}^{(j)} \sum_{\ell=1}^{L_+^{(j)}} \omega^{(\ell,j)} p_{D,k}^{(o)}(\mathbf{m}^{(\ell,j)}; \mathcal{S}^{(o)}) q_+^{(\ell,j)}(\mathbf{z}_k^{(o)}) & Z_k^{(o)} = \{\mathbf{z}_k^{(o)}\}, \end{cases} \quad (62)$$

under the same zeroth-order Taylor expansion of the detection model about the means.

The single sensor update operator is applied successively on the prior mode probability and the mode-conditioned prior $p_{k|k-1}^{(j)}(\mathbf{x}_k)$ for each observer resulting in the posterior mode probability $\mu_k^{(j)}$ and the mode-conditioned posterior

$$p_k^{(j)}(\mathbf{x}_k) \approx \sum_{\ell=1}^{L_k^{(j)}} \omega_k^{(\ell,j)} \mathcal{N}(\mathbf{x}_k; \mathbf{m}_k^{(\ell,j)}, \mathbf{P}_k^{(\ell,j)}). \quad (63)$$

Closed form solutions of the integrals in (45)–(46) and (55)–(57) are not available when the dynamics and measurement functions are nonlinear, so linearized approximations must be made. For outputting purposes, the posterior state distribution is found through the total probability theorem and is given by

$$p_k(\mathbf{x}_k | Z_{0:k}) = \sum_{j=1}^M \mu_k^{(j)} p_k^{(j)}(\mathbf{x}_k). \quad (64)$$

3.4 Gaussian Mixture Splitting

In problems with nonlinear and chaotic dynamics and nonlinear measurements, the linearized approximations of (45)–(46) and (55)–(57) may only be accurate within small neighborhoods of the component means. Care must then be taken to satisfy the key GM filter assumption that component covariances are commensurately small to ensure linearity within the local support. Adaptive GM filtering addresses these concerns by dynamically increasing the number of components in regions where linear assumptions are most strongly violated. Any component of the GM that violates the assumptions is approximated itself by an R -component GM [6–8].

This paper defines splitting in a new compact form as the operator $\tilde{G}_c$, which operates on an arbitrary multivariate GM $p(\mathbf{x})$ as

$$\tilde{G}_c[p(\mathbf{x})] = \sum_{i=1}^I G_c[\omega^{(i)} \mathcal{N}(\mathbf{x}; \mathbf{m}^{(i)}, \mathbf{P}^{(i)})], \quad (65)$$

where

$$G_c[\omega^{(i)} \mathcal{N}(\mathbf{x}; \mathbf{m}^{(i)}, \mathbf{P}^{(i)})] = \begin{cases} \omega^{(i)} \mathcal{N}(\mathbf{x}; \mathbf{m}^{(i)}, \mathbf{P}^{(i)}) & c(\omega^{(i)}, \mathbf{m}^{(i)}, \mathbf{P}^{(i)}) \leq 0, \\ \sum_{\ell=1}^R G_c[\omega^{(i,\ell)} \mathcal{N}(\mathbf{x}; \mathbf{m}^{(i,\ell)}, \mathbf{P}^{(i,\ell)})] & \text{otherwise,} \end{cases} \quad (66)$$

and $c(\omega^{(i)}, \mathbf{m}^{(i)}, \mathbf{P}^{(i)})$ is a splitting criterion. The operator is applied recursively on all newly generated components until no components satisfy the splitting criterion. Splitting is performed efficiently by utilizing a pre-generated library of optimal split parameters for the univariate standard Gaussian,

$$q(x) = \mathcal{N}(x; 0, 1) \approx \tilde{q}(x) = \sum_{\ell=1}^R \tilde{\omega}^{(\ell)} \mathcal{N}(x; \tilde{m}^{(\ell)}, \tilde{\sigma}^2) \quad (67)$$

as first proposed in [6] and refined in [7, 8]. By this approach, the univariate split parameters $\{\tilde{\omega}^{(\ell)}, \tilde{m}^{(\ell)}, \tilde{\sigma}\}_{\ell=1}^R$ can be referenced at runtime and applied to arbitrary multivariate Gaussian distributions via scaling, shifting, and covariance diagonalization.

Splitting can be performed in an arbitrary direction $\mathbf{u}$ while preserving the original component covariance [52]. The multivariate GM component can be split along the unit vector $\mathbf{u}$ as

$$\omega^{(i)} \mathcal{N}(\mathbf{x}; \mathbf{m}^{(i)}, \mathbf{P}^{(i)}) \approx \sum_{\ell=1}^R \omega^{(i,\ell)} \mathcal{N}(\mathbf{x}; \mathbf{m}^{(i,\ell)}, \mathbf{P}^{(i,\ell)}), \quad (68)$$

where

$$\omega^{(i,\ell)} = \tilde{\omega}^{(\ell)} \omega^{(i)}, \quad (69)$$

$$\mathbf{m}^{(i,\ell)} = \mathbf{m}^{(i)} + \tilde{m}^{(\ell)} \sigma_u^{(i)} \mathbf{u}, \quad (70)$$

$$\mathbf{P}^{(i,\ell)} = \mathbf{P}^{(i)} - (\sigma_u^{(i)})^2 \mathbf{u} \mathbf{u}^\top \sum_{j=1}^R (\tilde{m}^{(j)})^2 \tilde{\omega}^{(j)}, \quad (71)$$

$$\sigma_u^{(i)} = \frac{1}{\sqrt{\mathbf{u}^\top (\mathbf{P}^{(i)})^{-1} \mathbf{u}}}. \quad (72)$$

A full single step of the AGMIMM filter is summarized in Algorithm 1. The algorithm is presented for the general case of multiple sensors and reduces to the single-sensor AGMIMM for the case of $O = 1$.

Algorithm 1 Single AGMIMM step

Require: Z_k and $\mu_{k-1}^{(i)}, p_k^{(i)}(\mathbf{x}_{k-1}) \forall i \in \mathcal{M}$
for each $j \in \mathcal{M}$ **do**
 Compute $\bar{\mu}_{k|k-1}^{(j)}$ according to (27)
 Compute $\{\omega_{k-1|k-1}^{(\ell,j)}, \mathbf{m}_{k-1|k-1}^{(\ell,j)}, \mathbf{P}_{k-1|k-1}^{(\ell,j)}\}_{\ell=1}^L$ with GMRC algorithm [45]
 Compute $\{\omega_{k|k-1}^{(\ell,j)}, \mathbf{m}_{k|k-1}^{(\ell,j)}, \mathbf{P}_{k|k-1}^{(\ell,j)}\}_{\ell=1}^L$ according to (44)–(46)
 Initialize multi-sensor posterior as prior: $p^{(j)}(\mathbf{x}) \leftarrow p_{k|k-1}^{(j)}(\mathbf{x}_k)$
 Recursively split according to criterion c : $p^{(j)}(\mathbf{x}) \leftarrow \tilde{G}_c[p^{(j)}(\mathbf{x})]$
 Initialize prior mode probability as prior: $\bar{\mu}^{(j)} \leftarrow \bar{\mu}_{k|k-1}^{(j)}$
 for $o = 1, \dots, O$ **do**
 Apply single-sensor update: $p_+^{(j)}(\mathbf{x}) \leftarrow [\Psi_k^{(o,j)} p^{(j)}](\mathbf{x})$
 Recursively split according to criterion c : $p_+^{(j)}(\mathbf{x}) \leftarrow \tilde{G}_c[p_+^{(j)}(\mathbf{x})]$
 $p^{(j)}(\mathbf{x}) \leftarrow p_+^{(j)}(\mathbf{x})$
 Apply single-sensor update: $\bar{\mu}_+^{(j)} \leftarrow [\Psi_k^{(o,j)} \bar{\mu}^{(j)}]$
 $\bar{\mu}^{(j)} \leftarrow \bar{\mu}_+^{(j)}$
 end for
 $p_k^{(j)}(\mathbf{x}_k) \leftarrow p^{(j)}(\mathbf{x})$
 $\mu_k^{(j)} \leftarrow \bar{\mu}^{(j)}$
 Recursively split according to criterion c : $p_k^{(j)}(\mathbf{x}_k) \leftarrow \tilde{G}_c[p_k^{(j)}(\mathbf{x}_k)]$
end for
Compute state pdf for output: $p_k(\mathbf{x}_k|Z_{0:k}) \leftarrow \sum_{j=1}^M \mu_k^{(j)} p_k^{(j)}(\mathbf{x}_k)$

4 Multi-sensor Maneuvering Cislunar Space Object Tracking

This section presents the application of the novel AGMIMM filter to the problem of tracking a maneuvering CSO using angles-only data from space-based observers. Section 4.1 defines a splitting criterion based on statistical linear regression that accounts for errors resulting from the linearization of nonlinear dynamics or measurement functions. Section 4.2 presents a recently proposed splitting criterion for the variance of the Jacobi constant. This criterion aims to combat covariance growth through the CR3BP dynamics as a result of their chaotic nature. Section 4.3 introduces the addition of negative information to utilize non-observations. Section 4.4 adds user-defined constraints on the possible Jacobi constant values and an assumption that the target does not crash into the Moon. Section 4.5 defines the target modes and develops an approximation of the discrete-time process noise from the assumed continuous-time noise values. Finally, Sect. 4.6 presents a numerically stable propagation scheme of the CR3BP equations.

4.1 Statistical Linearization Error Criterion

This paper adopts a statistical linearization approach, where the dynamics and measurement processes are linearized using deterministic weighted sample points. The statistical linear regression approach, which encompasses, for example, unscented, central differencing, and cubature filtering, employs the linear approximation

$$\mathbf{y} = \mathbf{g}(\mathbf{x}) \approx \mathbf{G}\mathbf{x} + \mathbf{b}, \quad (73)$$

where the generic nonlinear transformation $\mathbf{g}(\cdot) : \mathbb{R}^{n_x} \rightarrow \mathbb{R}^{n_y}$ can be replaced by the nonlinear solution flow $\mathbf{f}(\cdot)$ or measurement equation $\mathbf{h}(\cdot)$, and where $\mathbf{G}$ and $\mathbf{b}$ are optimized to minimize the sum of the squares of the linearization error

$$\mathbf{e} = \mathbf{g}(\mathbf{x}) - (\mathbf{G}\mathbf{x} + \mathbf{b}). \quad (74)$$

The statistical linearization approach is particularly attractive for AGM filtering, as the ensemble approximation of the linearization error covariance

$$\mathbb{E}[\mathbf{e}\mathbf{e}^\top] = \mathbf{P}_e \quad (75)$$

is readily available. With this, the splitting criterion of [53] is adopted, given by

$$c_{\text{SL}}(\omega, \mathbf{m}, \mathbf{P}; \mathbf{g}, s_{\max}, \Gamma) = s - s_{\max}, \quad (76)$$

$$s = \omega^\Gamma \cdot (1 - \exp(-\epsilon))^{1-\Gamma} \in [0, 1], \quad (77)$$

$$\epsilon = \text{tr}(\mathbf{P}_e), \quad (78)$$

where Γ and $s_{\max}$ are design parameters and s represents a geometric interpolation of the component weight and linearization error. The tuning parameter $\Gamma \in [0, 1]$ denotes the relative importance of component weight and linearization error in the calculation of the selection criterion; when $\Gamma = 0$, only linearization error is considered, and when $\Gamma = 1$, only component weight is considered [53]. If, for a given component, s exceeds a user-defined threshold $s_{\max}$, the component is split.

Components that are marked for splitting by this criterion are split along the direction of highest nonlinearity. To measure the nonlinearity of $\mathbf{g}(\mathbf{x})$ in a particular direction $\mathbf{u}$, this paper adopts the squared Frobenius norm of the directional derivative

$$\|\nabla_{\mathbf{u}} \mathcal{G}(\mathbf{x})|_{\mathbf{x}}\|_F^2 = \text{tr}\left(\nabla_{\mathbf{u}} \mathcal{G}(\mathbf{x})|_{\mathbf{x}} (\nabla_{\mathbf{u}} \mathcal{G}(\mathbf{x})|_{\mathbf{x}})^\top\right) = \mathbf{u}^\top \left(\sum_{i=1}^{n_y} \mathbf{D}_i^\top(\mathbf{x}) \mathbf{D}_i(\mathbf{x}) \right) \mathbf{u} \quad (79)$$

as proposed in [52], where the Jacobian $\mathcal{G}(\mathbf{x}) = \nabla_{\mathbf{x}} \mathbf{g}(\mathbf{x})$ and the Hessian $\mathbf{D}_i(\mathbf{x}) = \nabla_{\mathbf{x}} \nabla_{\mathbf{x}} g_i(\mathbf{x})|_{\mathbf{x}}$. By this nonlinearity measure, the constrained optimal direction is

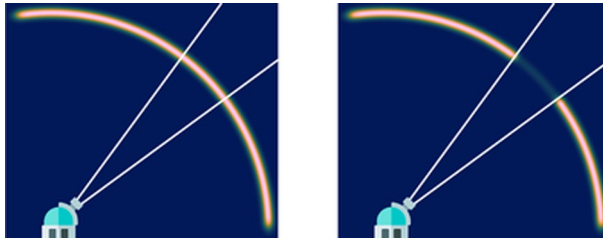


Fig. 3 Growth in covariance as a result of different Jacobi constant values where black denotes periodic halo orbits, dark blue denotes the initial points and their covariance, and light blue denotes the propagated points and their covariance

$$\mathbf{u}^* = \arg \max_{\mathbf{u}^T \mathbf{P}^{-1} \mathbf{u} = 1} \left\{ \mathbf{u}^T \left(\sum_{i=1}^{n_y} \mathbf{D}_i^T(\mathbf{x}) \mathbf{D}_i(\mathbf{x}) \right) \mathbf{u} \right\}, \quad (80)$$

where the constraint $\mathbf{u}^T \mathbf{P}^{-1} \mathbf{u}$ limits $\mathbf{u}$ to the 1σ ellipsoid of $\mathbf{P}$ and ensures that both the nonlinearity and uncertainty are considered. Through the whitening change of variables $\mathbf{y} = \mathbf{S}^{-1} \mathbf{u}$, (80) can be transformed to

$$\mathbf{u}^* = \mathbf{S} \mathbf{y}^*, \quad \mathbf{y}^* = \arg \max_{\|\mathbf{y}\|=1} \left\{ \mathbf{y}^T \mathbf{S}^T \left(\sum_{i=1}^{n_y} \mathbf{D}_i^T(\mathbf{x}) \mathbf{D}_i(\mathbf{x}) \right) \mathbf{S} \mathbf{y} \right\}, \quad (81)$$

$$\equiv \arg \max_{\|\mathbf{y}\| \neq 0} \left\{ \frac{\mathbf{y}^T \mathbf{S}^T \left(\sum_{i=1}^{n_y} \mathbf{D}_i^T(\mathbf{x}) \mathbf{D}_i(\mathbf{x}) \right) \mathbf{S} \mathbf{y}}{\mathbf{y}^T \mathbf{y}} \right\}. \quad (82)$$

References [9, 52] show that, by the Rayleigh-Ritz inequality, $\mathbf{y}^*$ is the eigenvector corresponding to the maximum eigenvalue of $\mathbf{S}^T \left(\sum_{i=1}^{n_y} \mathbf{D}_i^T(\mathbf{x}) \mathbf{D}_i(\mathbf{x}) \right) \mathbf{S}$.

4.2 Jacobi Constant Criterion

In addition to the linearization error splitting, the AGMIMM filter employs a newly defined splitting criterion based on Jacobi constant values [4]. By definition, the value of the Jacobi constant along a given ballistic trajectory remains constant. Differences in Jacobi constant between two solutions thus reflect different orbits/trajectories, whereas neighboring solutions of the same Jacobi constant likely belong to the same trajectory but separated by phase. Thus, the Jacobi constant variance serves as a predictor of linearization error [4]. This relationship between linearization error and Jacobi constant variance is demonstrated in Fig. 3, where the initial regression point ensemble covariance grows significantly over time. If, prior to propagation, a given component's Jacobi constant variance exceeds a user-defined threshold, the component is split.

The Jacobi constant variance associated with the component $(\omega, \mathbf{m}, \mathbf{P})$ is given by

$$\sigma_C^2 = \int (C(\mathbf{r}, v) - E[C(\mathbf{r}, v)])^2 \mathcal{N}(\mathbf{x}; \mathbf{m}, \mathbf{P}) d\mathbf{x}, \quad (83)$$

where the inner expectation $E[\cdot]$ is also taken with respect to $\mathcal{N}(\mathbf{x}; \mathbf{m}, \mathbf{P})$. Equation (83) can be trivially obtained using the already computed regression points. The associated split criterion function is then defined as

$$c_{JC}(\omega, \mathbf{m}, \mathbf{P}; \sigma_{C, \max}^2, \omega_{\min}) = \min \left(\sigma_C^2 - \sigma_{C, \max}^2, \omega - \omega_{\min} \right) \quad (84)$$

such that a component is split if its GM weight is larger than the minimum allowable value $\omega_{\min}$ and its associated Jacobi constant variance σ_C^2 exceeds the user-specified threshold $\sigma_{C, \max}^2$. Components marked for splitting by this criterion are split along the direction of highest nonlinearity of the dynamics function as in (81).

While the Jacobi constant variance is not a direct measure of nonlinearity, its computational cost is far cheaper than other measures of nonlinearity that require numerous and expensive computations of the state transition matrix. Because selection criterion tests must be performed more frequently than split direction optimization, the Jacobi constant variance criterion is used only in the selection of components to improve the filter execution speed. Then, for the subset of components selected for splitting, the full nonlinearity measure is used to determine the best split direction.

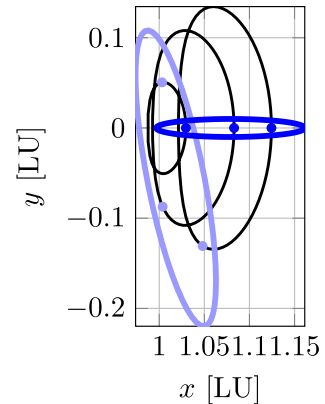
4.3 Splitting for Negative Information

In the cislunar regime, long periods of multiple sensing opportunities are rare due to solar illumination, Earth/Moon occlusion, and the Moon exclusion angle. It is important to incorporate any information gained from not detecting a target, termed negative information. Negative information is incorporated into the posterior pdf via the probability of detection function $p_{D,k}^{(o)}(\mathbf{x}_k; \mathcal{S}^{(o)})$. Note that the expansion approximation applied in (50) and (51) effectively assumes the probability of detection of a single component to be constant over its local support. Thus, significant approximation errors may arise when large components overlap the sensor FoV boundaries due to a sharp change in the probability of detection within the component domain. To address these discontinuities, the proposed filter employs a geometric recursive splitting algorithm that splits components that overlap FoV bounds until some stopping criteria are met [11, 12], as illustrated in Fig. 4. The details of this algorithm and its associated criterion function c are omitted for brevity but can be found in [12].

4.4 Nonlinear Constraints

The proposed filter provides the ability to add any additional nonlinear constraints or soft data from human sources. For example, with knowledge of the target's ΔV capacity, bounds can be imparted on the Jacobi constant as

Fig. 4 Example space object state probability density function before (left) and after (right) incorporating negative information (non-detection), obtained using recursive Gaussian mixture splitting [4]



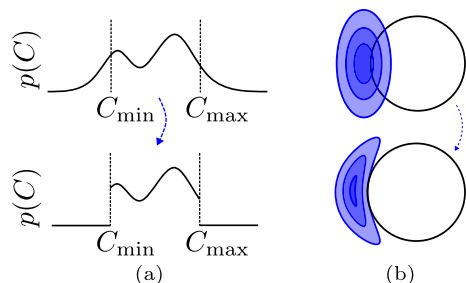
$$C_{\min}(\mathbf{r}_0, v_0 + \Delta V) \leq C(\mathbf{r}, v) \leq C_{\max}(\mathbf{r}_0, v_0 - \Delta V), \quad (85)$$

where the upper bound is valid under the assumption $v_0 \geq \Delta V$. To incorporate this constraint, components that violate the Jacobi constant bounds can be pruned from the GM. In the context of tracking a maneuvering CSO where no knowledge of the ΔV capacity is available, a conservative minimum and maximum can be chosen that account for the maximum realistic ΔV capacity in cislunar operations. While exceeding the Jacobi constant bounds implies that a particular ΔV has been exceeded, the converse is not necessarily true. For example, a spacecraft performing routine station-keeping maneuvers to can expend its ΔV capacity over time without significantly changing its orbit's Jacobi constant. Additionally, the assumption of target-lunar collision avoidance can be incorporated into the posterior. When components are in the Moon's vicinity, they can be split in a similar manner to the application of negative information. The portion of the density within the Moon can then be truncated. These nonlinear constraints can be seen graphically in Fig. 5.

4.5 Target Modality Definitions

In a nonlinear JMS system, modes are used to differentiate between varying dynamics, measurement models, and/or noise distributions. To account for a maneuvering

Fig. 5 Graphical depiction of the nonlinear constraints for **a** Jacobi constant bounds and **b** Moon collision



CSO, this paper assumes mode-independent dynamics, measurements, and measurement noise covariance and models unknown maneuvers via a mode dependent process noise covariance. In particular, the proposed filter has two modes, where ballistic trajectories are denoted by $\tau_k = 1$ and maneuvers are denoted by $\tau_k = 2$. The continuous-time process noise covariance for a given mode is defined as

$$\mathcal{Q}^{(j)} = E[\mathbf{w}(t, \tau = j)\mathbf{w}(t, \tau = j)^\top] = \begin{bmatrix} (q_{w_x}^{(j)})^2 & 0 & 0 \\ 0 & (q_{w_y}^{(j)})^2 & 0 \\ 0 & 0 & (q_{w_z}^{(j)})^2 \end{bmatrix}. \quad (86)$$

Following standard results (see, e.g. [54, Chap. 4]), the discrete-time process noise covariance mapped to the state space is found via the solution of

$$\mathbf{Q}_{k-1}^{(j)} = \int_{t_{k-1}}^{t_k} \Phi(t_k, t) \Gamma \mathcal{Q}^{(j)} \Gamma^\top \Phi^\top(t_k, t) dt, \quad (87)$$

where $\Phi(t_k, t)$ is the state transition matrix from t to t_k and

$$\Gamma = \begin{bmatrix} \mathbf{0}_{3 \times 3} \\ \mathbf{I}_{3 \times 3} \end{bmatrix}. \quad (88)$$

Equation (87) does not have a closed-form solution for the system presented in this paper, and thus, the discrete-time process noise covariance is found through the coupled numerical integration of (6) and

$$\dot{\mathbf{Q}}^{(j)} = \mathbf{F} \mathbf{Q}^{(j)} + \mathbf{Q}^{(j)} \mathbf{F}^\top + \Gamma \mathcal{Q}^{(j)} \Gamma^\top \quad (89)$$

over $t \in [t_{k-1}, t_k]$, where the initial condition $\mathbf{Q}_{k-1}^{(j)} = \mathbf{0}_{6 \times 6}$, and $\mathbf{F}$ is the Jacobian of the continuous-time dynamics given by

$$\mathbf{F} = \begin{bmatrix} \mathbf{0}_{3 \times 3} & \mathbf{I}_{3 \times 3} \\ \frac{\partial^2 U}{\partial x^2} & \frac{\partial^2 U}{\partial x \partial y} & \frac{\partial^2 U}{\partial x \partial z} \\ \frac{\partial^2 U}{\partial y \partial x} & \frac{\partial^2 U}{\partial y^2} & \frac{\partial^2 U}{\partial y \partial z} \\ \frac{\partial^2 U}{\partial z \partial x} & \frac{\partial^2 U}{\partial z \partial y} & \frac{\partial^2 U}{\partial z^2} \end{bmatrix} \mathbf{B}, \quad \mathbf{B} = \begin{bmatrix} 0 & 2 & 0 \\ -2 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (90)$$

4.6 Regularized Dynamics for Numerically Stable Propagation

The CR3BP EOMs include singularities at the positions of the Earth and Moon centers of mass, which introduce numerical instability when propagating target motion through close approaches. Accurate numerical integration of the EOMs in the vicinity of the primaries is essential for maintaining track custody of a CSO during periods of long observation gaps. As such, a process of regularization [55–58] is employed to provide increased numerical stability during close approaches.

The process of regularization involves the introduction of a fictitious time γ and a coordinate transformation from the 6-dimensional state to an 8-dimensional state. Through the application of regularization, it is possible to remove the singularity at the Earth or Moon. The work presented in this paper considers only close approaches of the Moon, and thus, removes the singularity at the Moon's location as formulated in [55] and summarized below.

The fictitious time is related to the simulation time t via the differential equation

$$\frac{dt}{d\gamma} = r_{/Moon}. \quad (91)$$

By this definition, simulation time flows more slowly when compared to that of the fictitious time during close approaches of the Moon, thus mitigating numerical instability. Letting $(\cdot)'$ denote derivatives with respect to γ and remembering that the position of the Moon is constant, the unperturbed CR3BP EOMs are equivalently represented in vector form as

$$\mathbf{r}_{/Moon}'' = \frac{\mathbf{r}_{/Moon} \cdot \mathbf{r}_{/Moon}'}{\mathbf{r}_{/Moon} \cdot \mathbf{r}_{/Moon}} \mathbf{r}_{/Moon}' + r_{/Moon} \mathbf{B} \mathbf{r}_{/Moon}' - \mu \frac{\mathbf{r}_{/Moon}}{r_{/Moon}} + r_{/Moon}^2 \boldsymbol{\eta}, \quad (92)$$

where

$$\boldsymbol{\eta} = \left[x - \frac{(1-\mu)(x+\mu)}{r_{/Earth}^3} \quad y - \frac{(1-\mu)y}{r_{/Earth}^3} \quad -\frac{(1-\mu)z}{r_{/Earth}^3} \right]^\top. \quad (93)$$

It is clear to see that (92) still contains a singularity at the Moon which is rectified by the K-S Formulation [58], a generalization of the Levi-Civita Formulation. Through the state augmentation

$$\bar{\mathbf{r}}_{/Moon} = \begin{bmatrix} \mathbf{r}_{/Moon}^\top & 0 \end{bmatrix}^\top, \quad \mathcal{B} = \begin{bmatrix} \mathbf{B} & \mathbf{0}_{3 \times 1} \\ \mathbf{0}_{1 \times 3} & 0 \end{bmatrix}, \quad \bar{\boldsymbol{\eta}} = [\boldsymbol{\eta}^\top \quad 0]^\top \quad (94)$$

and the introduction of the nonlinear transformation

$$\bar{\mathbf{r}}_{/Moon} = \mathcal{L}(\bar{\mathbf{u}}) \bar{\mathbf{u}}, \quad \bar{\mathbf{u}} = [u_1 \quad u_2 \quad u_3 \quad u_4]^\top, \quad \mathcal{L}(\bar{\mathbf{u}}) = \begin{bmatrix} u_1 & -u_2 & -u_3 & u_4 \\ u_2 & u_1 & -u_4 & -u_3 \\ u_3 & u_4 & u_1 & u_2 \\ u_4 & -u_3 & u_2 & -u_1 \end{bmatrix} \quad (95)$$

the unperturbed CR3BP EOMs can be shown to be equivalent to

$$\bar{\mathbf{u}}'' - \frac{h}{2} \bar{\mathbf{u}} = \mathcal{L}^\top(\bar{\mathbf{u}}) \mathcal{B} \mathcal{L}(\bar{\mathbf{u}}) \bar{\mathbf{u}}' + \frac{r_{/Moon}}{2} \mathcal{L}^\top(\bar{\mathbf{u}}) \bar{\boldsymbol{\eta}}, \quad h = \frac{1-\mu}{r_{/Earth}} + \frac{x^2 + y^2}{2} - \frac{C}{2}. \quad (96)$$

While the computation of C retains a singularity at the Moon, its value is constant on a given ballistic trajectory. Thus, it can be computed prior to numerical integration resulting in no singularities at the Moon in (96). To compute the discrete-time process noise covariance concurrently, (89) is rewritten as

$$(\mathbf{Q}^{(j)})' = (\mathbf{F}\mathbf{Q}^{(j)} + \mathbf{Q}^{(j)}\mathbf{F}^\top + \mathbf{\Gamma}\mathbf{Q}^{(j)}\mathbf{\Gamma}^\top)r_{/\text{Moon}}. \quad (97)$$

In this work, numerical integration of (96) and (97) is performed using the explicit Runge–Kutta Dormand–Prince method [59] of order 5(4). The solution stability is verified by testing that the boundary point Jacobi constant values are within 10^{-12} of each other.

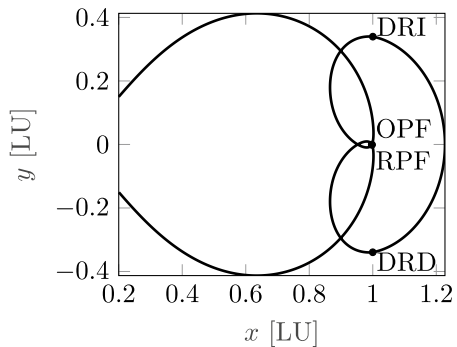
5 Results

The AGMIMM is first compared to the IMM and GMIMM in a simplified problem involving CR3BP dynamics and no measurements. In this test, the ability of the considered filters to accurately approximate the forecasted state pdf subject to chaotic dynamics and random mode switches is examined. A representative 20 component mixed mode-conditioned posterior density¹ is constructed and shown in Fig. 2a. The IMM, GMIMM, and AGMIMM approximations are then applied to the mixed mode-conditioned posterior, yielding the hypothesis-merged mixtures shown in Fig. 2b, d and f, respectively. The GMIMM and AGMIMM approximations are reduced to ten components, whereas the IMM approximation is Gaussian. As evidenced by the 1,000 overlaid Monte Carlo samples, the AGMIMM provides a better approximation of the true non-Gaussian pdf than the GMIMM for the same number of components. The 1σ curves of the components of the GMs are provided for clarity. Figure 2c, e, g show the resulting IMM, GMIMM and AGMIMM priors after a 21 day propagation and assumed ballistic model. While 21 day observation gaps may not be representative of future cislunar SDA operations, the example serves to highlight the peculiar non-Gaussian uncertainty structures produced by the multi-body dynamics and the AGMIMM filter's corresponding ability to capture them through adaptive splitting.

The AGMIMM filter is also tested against a simplified cislunar trajectory inspired by the Artemis I mission. Unlike the Artemis I trajectory, which has a total of 19 reported maneuvers [60], the simplified trajectory features four impulsive maneuvers designed to mimic the Artemis I outbound powered flyby (OPF), distant retrograde insertion (DRI), distant retrograde departure (DRD), and return powered flyby (RPF) maneuvers and is presented in Fig. 6. The initial epoch of the simulation is November 16, 2022, 08:44:51.150 UTC chosen to align with the timeline of the Artemis I mission, and the total simulation time is 28.30 days. The $\Delta\mathbf{v}$ magnitudes of the simplified OPF, DRI, DRD, and RPF maneuvers are 0.1980, 0.0997, 0.0997, and 0.1980 [km/s], respectively. Details of the maneuver times and corresponding changes in velocity with respect to the synodic frame are reported in Table 1. The initial state of the target is defined as

$$\mathbf{r}_0 = [75, 926.589 \ 56, 944.942 \ 0]^\top \text{ [km]}, \quad (98)$$

¹ Data publicly available at <https://gist.github.com/keithalegrand/51997ac197421d42e8059cb6a8fdb933>.

Fig. 6 Simplified Artemis I trajectory**Table 1** Impulsive maneuvers

Maneuver	$t - t_0$ [day]	$\Delta \mathbf{v}^T$ [km/s]
OPF	4.6488	$[-0.1385 \ -0.1415 \ 0]$
DRI	10.1894	$[-0.0878 \ 0.0472 \ 0]$
DRD	18.1152	$[-0.0878 \ -0.0472 \ 0]$
RPF	23.6558	$[-0.1385 \ 0.1415 \ 0]$

$$\mathbf{v}_0 = [1.5636 \ 1.8042 \ 0]^T \text{ [km/s]}, \quad (99)$$

with an initial position and velocity uncertainty of 20 [km] and 1 [m/s] root sum square (RSS), respectively. Between the discrete maneuvers, the true trajectory is propagated according to the continuous-time CR3BP EOMs. If t_k^- is the time immediately prior to an impulsive maneuver, the true post-maneuver position and velocity are modeled as $\mathbf{r}(t_k^+) = \mathbf{r}(t_k^-)$ and $\mathbf{v}(t_k^+) = \mathbf{v}(t_k^-) + \Delta \mathbf{v}$, respectively. The clear difference between the simulated true maneuver model and the filter's assumed white-noise process is, in fact, valuable in the evaluation of the AGMIMM's robustness to model mismatch.

A notional cislunar space-based surveillance network is considered that consists of three observer spacecraft in an Earth–Moon 1:1 planar resonant periodic orbit, as shown in Fig. 7, with initial conditions

$$\mathbf{r}_{0,\text{obs}}^{(1)} = [212, 171.440 \ 224, 189.013 \ 0]^T \text{ [km]}, \quad \mathbf{v}_{0,\text{obs}}^{(1)} = [0.1618 \ 0.5780 \ 0]^T \text{ [km/s]}, \quad (100)$$

$$\mathbf{r}_{0,\text{obs}}^{(2)} = [267, 266.589 \ 291, 378.691 \ 0]^T \text{ [km]}, \quad \mathbf{v}_{0,\text{obs}}^{(2)} = [0.4596 \ 0.1991 \ 0]^T \text{ [km/s]}, \quad (101)$$

$$\mathbf{r}_{0,\text{obs}}^{(3)} = [361, 478.978 \ 291, 915.604 \ 0]^T \text{ [km]}, \quad \mathbf{v}_{0,\text{obs}}^{(3)} = [0.5970 \ -0.1946 \ 0]^T \text{ [km/s]}. \quad (102)$$

A 1:1 resonant orbit is selected to mimic the nearly 1:1 behavior of the Sun in the synodic frame. As such, it is likely that a sensor with suitable lighting conditions at one epoch will have suitable lighting conditions at a later epoch. When visibility

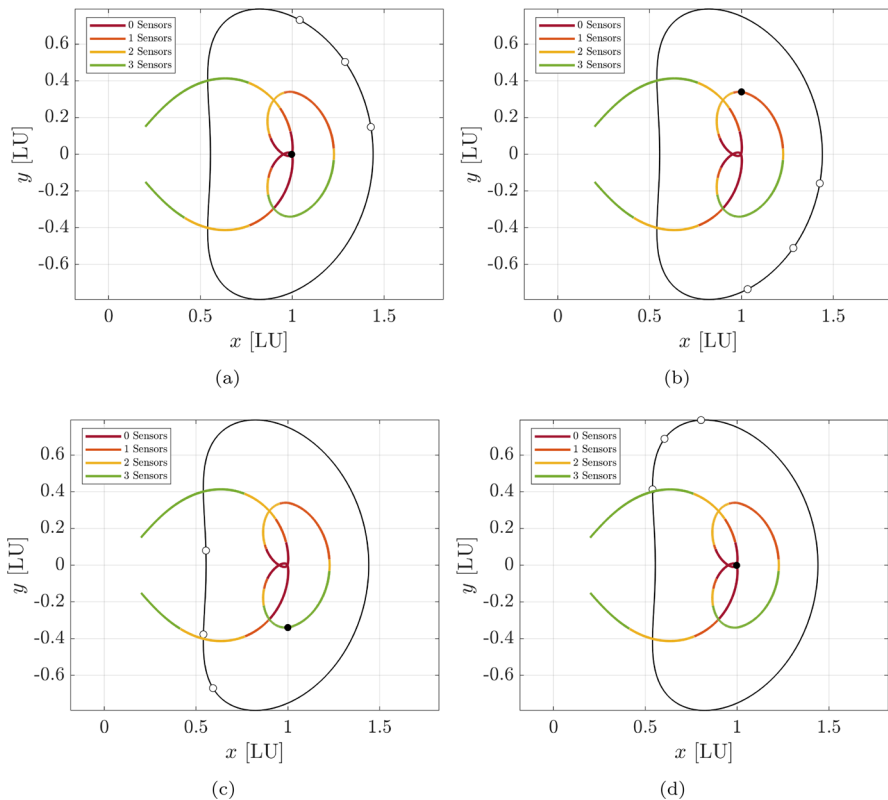


Fig. 7 Visibility of the target along the entire trajectory where colored lines denote the number of sensors that can measure the target at a given location, the black line denotes the sensors' orbit, the black dot denotes the location of the target, and black circles denote the locations of the sensors. The target and sensor geometry is shown for the **a** OPF, **b** DRI, **c** DRD, and **d** RPF maneuvers

conditions are satisfied, angle measurements are simulated according to (13) and corrupted by white Gaussian noise with a standard deviation of 18 arcsec, which reflects a single pixel error of a camera with a $10^\circ \times 10^\circ$ FoV and 2048×2048 pixel focal plane array. For ease of analysis, the sensor attitudes are designed to track the true object position. Lighting conditions are simulated based on historical solar ephemerides from the Artemis I mission period, and when available, measurements are taken every 240 s.

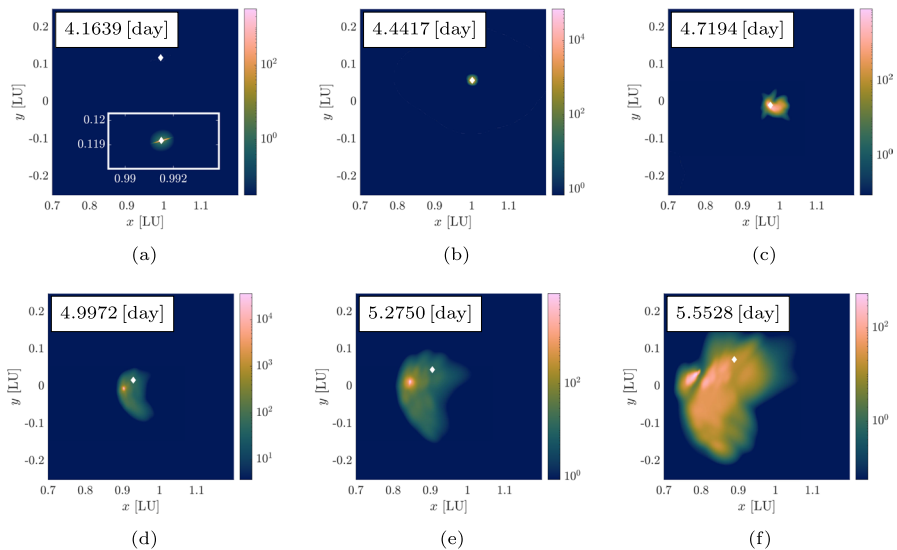
Figure 7 presents the number of sensors that can measure the target along its trajectory according to the illumination conditions, Earth/Moon occlusion, umbra, and Moon exclusion angle. The most impactful constraint is the Moon exclusion angle, which results in long observation gaps during which the OPF and RPF maneuvers take place. The two observations gaps last for 1.686 and 2.606 days, respectively. Other pertinent parameters of the AGMIMM are presented in Table 2.

During periods of high sensor coverage and high frequency measurements, the evolution of the state pdf is nearly Gaussian. On the other hand, periods of

Table 2 AGMIMM filter parameters

	Parameter	Value
Jump Markov parameters	π_{11}	0.99
	π_{12}	0.01
	π_{21}	0.1
	π_{22}	0.9
Gaussian splitting	L	100
	$s_{\max}$	0.1
	$\omega_{\min}$	0.01
	$\sigma_{C,\max}^2$	0.0001
	$C_{\min}$	1.75 [LU ² /TU ²]
	$C_{\max}$	3.75 [LU ² /TU ²]
	$q_{w_x}^{(1)}, q_{w_y}^{(1)}, q_{w_z}^{(1)}$	10 ⁻³ [LU ² /TU ³]
	$q_{w_x}^{(2)}, q_{w_y}^{(2)}, q_{w_z}^{(2)}$	1 [LU ² /TU ³]

measurement unavailability result in highly non-Gaussian densities and highlight the advantages of the AGMIMM filter. Figure 8 shows the AGMIMM outputs during the first observation gap. The x - y marginals of the state pdf are shown every 400 minutes over a 2000 minute window. At $t - t_0 = 4.1639$ [day], the density is approximately Gaussian with low covariance due to its conditioning on recent measurements. As time progresses, the nonlinear and dynamically-sensitive motion during the close Moon encounter results in a highly non-Gaussian evolution that is captured

**Fig. 8** Time evolution x - y marginal densities during first observation gap, which includes the target OPF maneuver, where the white diamond denotes the truth and the log color scale denotes values of the pdf

by the AGMIMM filter. Furthermore, the CSO performs the OPF maneuver at perilune ($t - t_0 = 4.6466$ [day]) and deviates from the ballistic trajectory. By the specified target-mode probabilities, the ballistic hypothesis is more likely, resulting in a distinct density peak corresponding to the ballistic forecast, as shown in Fig. 8d, e. Figure 8f shows the marginal density at $t - t_0 = 5.5528$ [day], at which time the CSO is still not visible to any of the space based observers. Despite the CSO's non-visibility, non-detections from the middle observer are processed, and the negative information content is used to refine the pdf. The impact of negative information is apparent in Fig. 8f, where the boundaries of the middle observer's FoV and lunar exclusion cone manifest as sharp linear features in the marginal pdf. Critically, this same negative information all but eliminates the possibility that the CSO was ballistic during the gap, even before a positive detection of the CSO is made.

For each of the considered filters, the state estimate is taken to be the posterior distribution mean $E[\mathbf{x}_k]$, and the estimation error at time t_k is

$$\mathbf{e}_k = \mathbf{x}_k - E[\mathbf{x}_k]. \quad (103)$$

The AGMIMM state estimation performance throughout the simulation is shown by the state estimation error magnitude and covariance 3σ bounds in Fig. 9. The estimation error varies by orders of magnitude depending on the sensor coverage. During periods of complete coverage, the position estimation error is often less than a kilometer in magnitude. Unsurprisingly, the largest estimation errors occur during long observation gaps and immediately following CSO maneuvers. The OPF and RPF maneuvers occur during the two observations gaps and thus correspond to the two largest errors and uncertainty. The DRI maneuver, however, occurs at a time when the CSO is visible to only one observer. Despite these unfavorable visibility conditions, the AGMIMM successfully retains custody of the CSO target during these maneuvers. Total sensor coverage is achieved at the time of the DRD maneuver, and

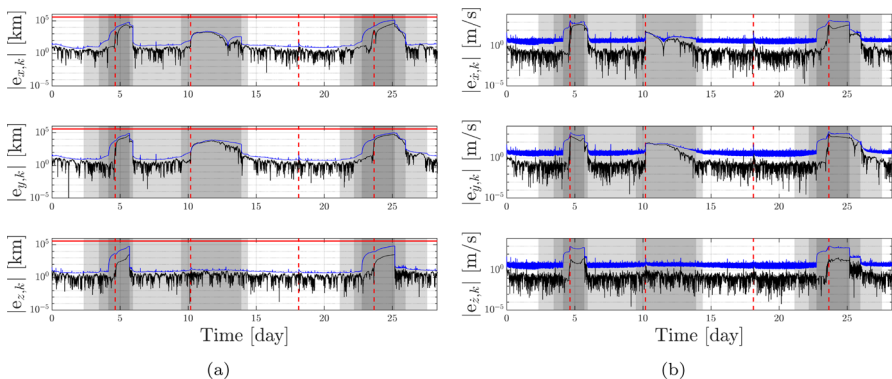


Fig. 9 AGMIMM **a** position and **b** velocity estimation performance, where black denotes the estimation error, blue denotes the 3σ bounds, the solid red lines denote the Earth–Moon semi-major axis, red dashed lines indicate maneuver times, and the shaded regions depict the level of sensor coverage, where darker regions indicate less sensor coverage

the estimation performance is indistinguishable from that of the ballistic trajectory arcs.

To evaluate the robustness and consistency of the AGMIMM, a Monte Carlo analysis is performed. The initial estimation error and measurement error sequences are randomly varied in each of the 250 trials. The filter is initialized with a Gaussian distribution with covariance $\mathbf{P}_0 = \text{diag}([\sigma_r^2 \ \sigma_r^2 \ \sigma_r^2 \ \sigma_v^2 \ \sigma_v^2 \ \sigma_v^2])$ where $\sigma_r = 20/\sqrt{3}$ [km] and $\sigma_v = 1/\sqrt{3}$ [m/s]. The initial mean is drawn from a Gaussian distribution centered at the true initial CSO state $\mathbf{x}(t_0) = [\mathbf{r}_0^T \ \mathbf{v}_0^T]^T$ and covariance $\mathbf{P}_0$.

The scaled normalized estimation error squared (SNEES) is computed at every time step and averaged over the 250 trials. Figure 10 presents the average SNEES (ASNEES) of the AGMIMM trials. The ASNEES is simply the average normalized estimation error squared (ANEES) [28] inversely scaled by the state dimension, such that an ASNEES value of one indicates a consistent filter, regardless of the state dimension [61]. ASNEES values significantly lower than one indicate that the filter is pessimistic or inefficient, whereas values significantly higher than one indicate that the filter is optimistic. For the majority of the simulation, the AGMIMM ASNEES hovers around 0.4, suggesting that the filter is consistent if not slightly conservative. The slight pessimism is due to modeling mismatch: the filter assumes that maneuvers can occur at any time step, whereas the four maneuver times were held constant in each trial to preserve the trajectory design. The same explanation applies to the dips in ASNEES seen before the OPF and RPF maneuvers, which are both unobserved. During the observation gap and before the maneuver, the ASNEES decreases due to the (appropriate) increase in estimation uncertainty reflecting the possibility of a maneuver. After the maneuver occurs, the ASNEES is restored as the estimation error increases to a level commensurate to the reported covariance. Brief spikes in the ASNEES are observed at the DRI and DRD maneuver times and correspond to the sudden changes in the true velocity, which are modeled as instantaneous burns for simplicity.

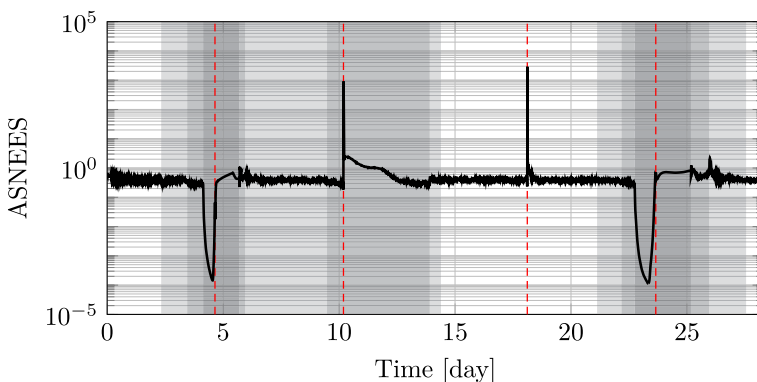


Fig. 10 ASNEES of the AGMIMM, where red dashed lines indicate maneuver times and the shaded regions depict the level of sensor coverage, where darker regions indicate less sensor coverage

To compare the AGMIMM to other state of the art filters, the 250 Monte Carlo trials are repeated using the square-root unscented Kalman filter, SRUKF [62], IMM [29], and GMIMM [30] filters. Because the SRUKF is inherently a single-model filter, the thrusting mode process noise covariance is used. While the AGMIMM maintains track custody of the CSO in all 250 trials, all trials of the SRUKF, IMM, and GMIMM fail shortly after the first observation gap. Here, failure is declared when the measurement likelihood agreement falls below the machine's smallest positive normalized floating-point number. Figure 11 presents the time history of the position error averaged over all of the trials, where failing trials are omitted from the average and the curve terminates when all trials have failed. The SRUKF exhibits the largest position error during the first period of full sensor coverage as well as the fastest error growth at the beginning of first observation gap. The IMM exhibits a lower error compared to that of the SRUKF when measurements are available. Up to the first (OPF) maneuver, the GMIMM and AGMIMM exhibit comparable position estimation error performance. During the dynamically sensitive OPF, the AGMIMM's adaptive and splitting and merging steps enable better approximation of the pdf evolution compared to the GMIMM, and the resulting estimation errors are around three orders of magnitude smaller. By the end of the first observation gap, only the AGMIMM maintains track custody, and continues to do so for the remainder of the simulation. Because the IMM, GMIMM, and AGMIMM are all nonlinear estimators and incorporate multiple models, the superior estimation performance of the AGMIMM is ascribed to its adaptive mixture approximation capability.

6 Conclusion

This paper presents a novel adaptive Gaussian mixture (AGM) filter that is uniquely able to track a maneuvering, noncooperative target in a highly nonlinear, chaotic system. The proposed adaptive Gaussian mixture interacting multiple model

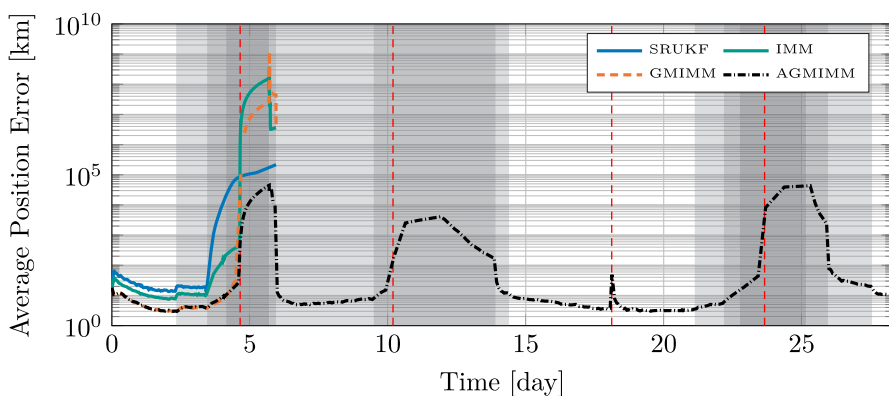


Fig. 11 Average position error, where red dashed lines indicate maneuver times and the shaded regions depict the level of sensor coverage, where darker regions indicate less sensor coverage

(AGMIMM) filter is suitable for nonlinear jump Markov systems (JMSs) where the target modality evolves according to a homogeneous Markov chain. Cluster-based merging is employed to maintain computational tractability while retaining accurate descriptions of the non-Gaussian state uncertainty. Furthermore, the AGMIMM employs multi-criterion parametric adaptation to account for nonlinear effects due to dynamics, measurements, negative information, and known constraints. In a 250-trial Monte Carlo cislunar space object (CSO) tracking simulation, the AGMIMM filter significantly outperforms existing maneuvering target tracking algorithms in terms of state estimation accuracy and robustness.

Appendix: Gaussian Mixture Reduction via Clustering

The Gaussian mixture reduction via clustering (GMRC) algorithm is presented in Algorithm 2. As opposed to a random initialization, the GMRC is kick-started by a greedy Runnall's reduction solution in the Preprocessing step. The Clustering step is described in Algorithm 3. Cluster assignments are represented by the $J \times L$ assignment matrix C , where the element $C_{j,\ell} = 1$ if component j is associated with cluster ℓ and $C_{j,\ell} = 0$ otherwise. In each iteration, the original components are grouped according to their closest cluster center, where the distance is measured by the Kullback–Leibler divergence or some other suitable distance measure $d(\cdot, \cdot)$. For a given cluster assignment C , the resulting mixture barycenters are computed according to Algorithm 4. A description of the optional Refinement step is omitted for brevity but is available in [45, 46].

Algorithm 2 Gaussian Mixture Reduction via Clustering (GMRC) [45]

Require: $\{\tilde{\omega}^{(j)}, \tilde{\mathbf{m}}^{(j)}, \tilde{\mathbf{P}}^{(j)}\}_{j=1}^J, L$

$$\{\omega^{(\ell)}, \mathbf{m}^{(\ell)}, \mathbf{P}^{(\ell)}\}_{\ell=1}^L \leftarrow \text{Preprocessing}(\{\tilde{\omega}^{(j)}, \tilde{\mathbf{m}}^{(j)}, \tilde{\mathbf{P}}^{(j)}\}_{j=1}^J, L)$$

$$\{\omega^{(\ell)}, \mathbf{m}^{(\ell)}, \mathbf{P}^{(\ell)}\}_{\ell=1}^L \leftarrow \text{Clustering}(\{\tilde{\omega}^{(j)}, \tilde{\mathbf{m}}^{(j)}, \tilde{\mathbf{P}}^{(j)}\}_{j=1}^J, \{\omega^{(\ell)}, \mathbf{m}^{(\ell)}, \mathbf{P}^{(\ell)}\}_{\ell=1}^L)$$

$$\{\omega^{(\ell)}, \mathbf{m}^{(\ell)}, \mathbf{P}^{(\ell)}\}_{\ell=1}^L \leftarrow \text{Refinement}(\{\tilde{\omega}^{(j)}, \tilde{\mathbf{m}}^{(j)}, \tilde{\mathbf{P}}^{(j)}\}_{j=1}^J, \{\omega^{(\ell)}, \mathbf{m}^{(\ell)}, \mathbf{P}^{(\ell)}\}_{\ell=1}^L)$$

return $\{\omega^{(\ell)}, \mathbf{m}^{(\ell)}, \mathbf{P}^{(\ell)}\}_{\ell=1}^L$

Algorithm 3 Clustering [45]

Require: $\{\tilde{\omega}^{(j)}, \tilde{\mathbf{m}}^{(j)}, \tilde{\mathbf{P}}^{(j)}\}_{j=1}^J, \{\omega^{(\ell)}, \mathbf{m}^{(\ell)}, \mathbf{P}^{(\ell)}\}_{\ell=1}^L$

$C \leftarrow \text{InitialClustering}(\{\tilde{\omega}^{(j)}, \tilde{\mathbf{m}}^{(j)}, \tilde{\mathbf{P}}^{(j)}\}_{j=1}^J, \{\omega^{(\ell)}, \mathbf{m}^{(\ell)}, \mathbf{P}^{(\ell)}\}_{\ell=1}^L)$

$\{\omega^{(\ell)}, \mathbf{m}^{(\ell)}, \mathbf{P}^{(\ell)}\}_{\ell=1}^L \leftarrow \text{ComputeCenters}(\{\tilde{\omega}^{(j)}, \tilde{\mathbf{m}}^{(j)}, \tilde{\mathbf{P}}^{(j)}\}_{j=1}^J, C)$

for $j = 1, \dots, J$ **do**

for $\ell = 1, \dots, L$ **do**

$C_{\text{tmp}} \leftarrow$ reassociate component $\{\tilde{\omega}^{(j)}, \tilde{\mathbf{m}}^{(j)}, \tilde{\mathbf{P}}^{(j)}\}$ to cluster ℓ

$\{\omega_{\text{tmp}}^{(\ell)}, \mathbf{m}_{\text{tmp}}^{(\ell)}, \mathbf{P}_{\text{tmp}}^{(\ell)}\}_{\ell=1}^L \leftarrow \text{ComputeCenters}(\{\tilde{\omega}^{(j)}, \tilde{\mathbf{m}}^{(j)}, \tilde{\mathbf{P}}^{(j)}\}_{j=1}^J, C_{\text{tmp}})$

$d_\ell \leftarrow d(\{\tilde{\omega}^{(j)}, \tilde{\mathbf{m}}^{(j)}, \tilde{\mathbf{P}}^{(j)}\}_{j=1}^J, \{\omega_{\text{tmp}}^{(\ell)}, \mathbf{m}_{\text{tmp}}^{(\ell)}, \mathbf{P}_{\text{tmp}}^{(\ell)}\}_{\ell=1}^L)$

end for

$\ell^* \leftarrow \arg \min_\ell d_\ell$

$C \leftarrow$ reassociate component $\{\tilde{\omega}^{(j)}, \tilde{\mathbf{m}}^{(j)}, \tilde{\mathbf{P}}^{(j)}\}$ to cluster ℓ^*

$\{\omega^{(\ell)}, \mathbf{m}^{(\ell)}, \mathbf{P}^{(\ell)}\}_{\ell=1}^L \leftarrow \text{ComputeCenters}(\{\tilde{\omega}^{(j)}, \tilde{\mathbf{m}}^{(j)}, \tilde{\mathbf{P}}^{(j)}\}_{j=1}^J, C)$

end for

return $\{\omega^{(\ell)}, \mathbf{m}^{(\ell)}, \mathbf{P}^{(\ell)}\}_{\ell=1}^L$

Algorithm 4 ComputeCenters [46]

Require: $\{\tilde{\omega}^{(j)}, \tilde{\mathbf{m}}^{(j)}, \tilde{\mathbf{P}}^{(j)}\}_{j=1}^J, C$

for $\ell = 1, \dots, L$ **do**

$\omega^{(\ell)} \leftarrow \sum_{C_{j,\ell}=1} \tilde{\omega}^{(j)}$

$\mathbf{m}^{(\ell)} \leftarrow \frac{1}{\omega^{(\ell)}} \sum_{C_{j,\ell}=1} \tilde{\omega}^{(j)} \tilde{\mathbf{m}}^{(j)}$

$\mathbf{P}^{(\ell)} \leftarrow \frac{1}{\omega^{(\ell)}} \sum_{C_{j,\ell}=1} \tilde{\omega}^{(j)} (\mathbf{P}^{(j)} + (\tilde{\mathbf{m}}^{(j)} - \mathbf{m}^{(\ell)})(\tilde{\mathbf{m}}^{(j)} - \mathbf{m}^{(\ell)})^\top)$

end for

return $\{\omega^{(\ell)}, \mathbf{m}^{(\ell)}, \mathbf{P}^{(\ell)}\}_{\ell=1}^L$

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Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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